Capability-Structured Agentic Multi-Agent Control of Conditional Handover and PDCP Duplication in LEO Satellite Networks

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***Abstract*—Low Earth orbit (LEO) satellite networks force user equipments (UEs) to change their serving satellites frequently. Conditional handover (CHO) and packet data convergence protocol (PDCP) duplication can protect service continuity during these changes, but they turn each handover into a sequence of coupled decisions on preparing a candidate, duplicating traffic, and switching, while many UEs compete for the same satellite resources under delayed and noisy telemetry. We formulate this joint control problem as a constrained decentralized partially observable Markov decision process and propose a capability-structured agentic framework based on centralized training with decentralized execution. The framework introduces perception, recurrent memory, ephemeris-based reasoning, congestion-price coordination, and primal–dual reliability regulation as separable capabilities, without placing a language model in the control loop. Across 25 independently trained policies, every learned policy–scenario pair achieves a lower outage ratio than a tuned 3GPP-inspired CHO rule, and the full agentic stack reduces the mean outage ratio by 60% under nominal conditions and by 46% under compound stress. Compared with a memoryless learned policy, the full stack attains statistically indistinguishable reliability with 39% fewer handover executions and a 70% lower ping-pong ratio. Targeted ablations identify goal regulation as the most influential single capability and show that the trained full-stack policies rely on both the congestion price and the price-aware reasoner, while a behavioral analysis reveals that the agents use CHO preparations as short-lived options, committing only 6% of them. We also identify the limitations of the approach, including increased preparation signalling and a higher sensitivity to compound disturbances.**

***Index Terms*—Agentic AI, conditional handover, PDCP duplication, LEO satellite networks, multi-agent reinforcement learning, non-terrestrial networks.**

I. Introduction

Non-terrestrial networks (NTNs) are expected to play a key role in sixth-generation (6G) wireless systems by extending broadband connectivity beyond the reach of terrestrial infrastructure [1]–[3]. Among NTN platforms, low Earth orbit (LEO) constellations are particularly attractive because their low altitude yields a shorter propagation delay and a lower path loss than geostationary systems while still covering large areas [1], [4]. The orbital motion that enables this coverage, however, continuously changes satellite visibility and limits the time for which any satellite can serve a given user equipment (UE), so that even stationary UEs must hand over frequently [4], [5]. Mobility management has therefore become a central design problem of LEO access networks.

Existing studies have addressed this problem mainly by deciding when to hand over and to which satellite. Classical schemes select the serving satellite according to the remaining visibility time, elevation angle, link quality, or load, and graph- or flow-based formulations coordinate multiple users under constellation-wide constraints [4], [6]. Location-aware conditional handover (CHO) exploits predictable satellite motion to trigger handovers in advance [7], AcceHo shortens handover latency by predicting future satellite access [8], SKYCASTLE mitigates service interruptions through distributed satellite anchors [9], and PMPS avoids handover interruptions by scheduling packets across several satellites when the terminal can track multiple beams [10]. These studies show that LEO mobility is predictable enough to be managed proactively rather than purely reactively.

Standardized protocols now provide the means for such proactive management. NR conditional reconfiguration allows a candidate target to be prepared before the serving link deteriorates and to be executed later [14], and packet data convergence protocol (PDCP) duplication allows the same packets to be sent over two lower-layer legs with explicit activation and deactivation [15]. Together with multi-connectivity, which exploits multiple radio paths for redundancy and reliability [11]–[13], these mechanisms let a UE prepare a target, protect its traffic temporarily with a second leg, and switch only when appropriate. As a result, a single handover decision becomes a sequence of preparation, duplication, and switching decisions.

Controlling this sequence well is considerably harder than selecting a target. Preparing candidates aggressively consumes scarce preparation resources, persistent duplication wastes radio capacity, and frequent switching causes unnecessary handovers and ping-pong, whereas acting too late exposes the UE to outage when its serving satellite sets. Moreover, the decisions of different UEs are coupled because they compete for the same satellite capacity and preparation resources, and each UE observes link quality and load only through delayed and noisy telemetry. A useful controller must therefore manage the whole lifecycle of each mobility procedure while accounting for other UEs and for imperfect information.

Learning-based control is a natural candidate for this task. DQN-based methods adapt handover decisions to changing network conditions [16], multi-agent deep reinforcement learning (RL) accounts for dynamic propagation, satellite capacity, and distributed user decisions [17], [18], and GEMS embeds spatial and temporal satellite mobility into dual-connectivity configuration [19], and hierarchical RL separates the decision of when to hand over from that of which satellite to select [20]. These approaches, however, still treat mobility mainly as

a choice of which satellite to use or when to hand over. The stateful interaction among preparation, duplication, switching, release, and rollback under shared resources and imperfect telemetry has thus remained largely unexplored.

Addressing this interaction requires capabilities beyond one-shot action selection. A controller must remember past observations to distinguish transient from persistent degradation, reason about the future availability of candidate satellites, coordinate with other UEs that compete for the same resources, and steer its behavior toward a reliability goal. These requirements correspond closely to the emerging notion of Agentic AI, which is characterized by autonomous perception, memory, reasoning, goal-directed action, and multi-agent coordination [21]–[23], and for which advanced deep RL has been identified as a key enabler in wireless networks [24]. Agentic AI thus offers a natural framework for structuring a mobility controller, provided that its realization fits the timescale of mobility control.

This condition matters because Agentic AI in communication networks does not require a large language model (LLM) to execute every real-time action. LLM-centric approaches suit tasks in which natural-language requirements must be interpreted, such as satellite spectrum orchestration [25], whereas AERS realizes decentralized agentic resource scheduling with autonomous beam agents that learn through interaction [26], and agentic multi-agent learning has also been applied to resource orchestration in satellite-assisted vehicular networks [27]. For protocol control at the timescale of a LEO control epoch, these examples suggest that agentic capabilities are best instantiated by lightweight, domain-specific mechanisms.

Following this principle, we propose a capability-structured agentic multi-agent framework for joint CHO and PDCP-duplication control in shared-capacity LEO networks. Each UE acts as an autonomous agent that decides when to prepare a candidate, activate or deactivate duplication, switch, release, or roll back, and all agents are trained by multi-agent proximal policy optimization (MAPPO) with centralized training and decentralized execution (CTDE). Recurrent memory summarizes local experience, an ephemeris-based reasoner anticipates the future viability of candidates, a scalar congestion price per satellite conveys shared-resource pressure without exchanging UE states or actions, and a primal–dual regulator adapts the reliability pressure during training. Because these capabilities are added one at a time, the resulting model family reveals not only whether learned control outperforms standardized rules but also how each capability changes the behavior of the agents.

The main contributions of this work are summarized as follows.

- We formulate joint CHO preparation, PDCP-duplication control, and primary-satellite switching in a shared-capacity LEO network as a constrained decentralized partially observable Markov decision process. The formulation captures delayed and noisy telemetry, time-varying satellite visibility, hidden link degradation, preparation contention, duplication overhead, and mobility cost.
- We develop a capability-structured CTDE-MAPPO framework in which perception, recurrent memory, ephemeris-based reasoning, and resource-aware coordination are introduced as distinct capabilities. This design supports capability-structured comparisons and targeted within-policy ablations without relying on a monolithic LLM-based controller.
- We introduce a lightweight coordination mechanism based on satellite congestion prices, which influences decentralized mobility decisions without UE-to-UE exchange of observations or actions, and a primal–dual regulator that replaces a fixed outage-penalty weight with an automatically updated multiplier; in our experiments, this regulator was the main source of the mobility savings of the learned policies.
- We evaluate the framework against conventional handover and duplication heuristics and a 3GPP-inspired A3/time-to-trigger CHO baseline, using independently trained capability variants, a goal-regulation-only variant, within-policy coordination and reasoning ablations, and compound traffic, capacity, and blockage disturbances. The evaluation characterizes not only service reliability but also duplication overhead, handover activity, ping-pong behavior, and shared-resource usage.

II. Related Work

A. LEO Mobility Management and Conditional Handover

The high orbital velocity of LEO satellites makes mobility management fundamentally different from that in terrestrial cellular networks. Satellite and user mobility jointly produce time-varying coverage, frequent handovers, and additional signaling and decision-making overhead [5]. Conventional LEO handover schemes therefore consider satellite-specific information such as elevation angle, remaining visibility time, channel availability, and load. For example, Zhang *et al.* formulate multi-user satellite handover as a network-flow problem whose edge weights reflect user requests and satellite service quality [6]. Juan *et al.* systematically examine 5G NR handover procedures in LEO networks and exploit predictable satellite motion and antenna information for mobility control [4].

The predictability of satellite motion has also motivated proactive handover mechanisms. Lee *et al.* propose location-based CHO for LEO NTN, using the relative distance between a UE and serving/target cell reference locations rather than relying only on received signal strength [7]. AcceHo reduces handover latency by predicting future satellite access and avoiding repeated interaction with the core network during the handover procedure [8]. At the network layer, SKYCAS-TLE deploys distributed satellite anchors and convergence-free forwarding to mitigate connection interruption and latency caused by the mobility of both users and LEO infrastructure [9]. These works establish the importance of proactive satellite mobility management, but they do not jointly determine candidate preparation, temporary PDCP duplication, and subsequent primary-link switching.

B. Multi-Connectivity and PDCP Duplication

Multi-connectivity enables a UE to simultaneously exploit radio resources from multiple access points and can improve

throughput, service continuity, and reliability through traffic aggregation or path redundancy [11]. Its potential in integrated terrestrial–satellite networks has also been demonstrated experimentally. Lopez *et al.* evaluate simultaneous terrestrial 5G and Starlink connectivity and show the potential of MC for reducing high-end latency and improving continuous service availability [12].

3GPP NTN studies explicitly consider multi-connectivity involving NTN-based NG-RAN and service continuity between NTN and terrestrial networks [13]. At the RRC layer, conditional reconfiguration enables candidate mobility configurations to be stored and conditionally executed [14]. At the user plane, the NR PDCP specification defines routing and duplication functions and supports activation/deactivation of PDCP duplication [15]. These standardized mechanisms provide the protocol building blocks for reliable mobility, but the standards intentionally leave many control decisions to network implementation.

Recent work has therefore studied how secondary links should be selected and maintained. GEMS formulates TN–NTN and NTN–NTN dual connectivity as a spatial matching problem and extends the graph temporally to account for future satellite mobility [19]. Such work demonstrates that continuity-aware MC must account for both current link utility and future service windows. Our problem is complementary but different: rather than determining only the dual-connectivity association, we control the *lifecycle* of the mobility procedure—preparation, duplication, commit, release, and rollback—under shared preparation and radio capacity.

Predictive multi-satellite scheduling goes one step further. PMPS exploits deterministic orbital motion to migrate traffic across the satellites that a multi-beam terminal can track simultaneously before the serving satellite departs, and it duplicates only the packets carried by satellites that are about to set, thereby replacing discrete handover interruptions with continuous multi-path operation [10]. This approach requires terminals with several concurrent beams and schedules packets rather than mobility procedures. In contrast, our work preserves the standardized single-serving-cell procedure and focuses on autonomously controlling its preparation, duplication, switching, and release decisions under shared satellite resources.

C. Learning-Based Satellite Mobility Control

The dynamics and large state spaces of satellite networks have motivated the use of RL for adaptive mobility management. Yang *et al.* develop the DQN-ALrM handover method for satellite–ground integrated networks, using a customized DQN with an adaptive learning rate to handle long-term multi-metric handover optimization [16]. Liu *et al.* consider mega-constellations under dynamic propagation conditions and develop centralized and distributed multi-agent deep-RL handover schemes that account for user-rate requirements and satellite load [17].

Distributed multi-agent control has also been investigated to reduce centralized signaling and computational overhead. Yang *et al.* formulate LEO handover as a decentralized decision process and employ multi-agent fingerprints with DDQN to mitigate training non-stationarity and load imbalance [18]. Zhao *et al.* decouple the handover decision into when to hand over and which satellite to select, and solve the two subproblems with a hierarchical framework that combines a GRU-based PPO agent for the timing with a DQN agent for the target selection [20]. GEMS uses graph attention to incorporate future satellite mobility into connectivity decisions [19]. Collectively, these studies demonstrate that learning-based policies can outperform static or myopic handover rules in dynamic environments.

However, existing learning-based mobility solutions predominantly treat handover as a serving-target or trigger decision. A CHO and PDCP-duplication procedure instead has an internal protocol state: a target may first be prepared, duplication may subsequently be activated or deactivated, a switch may be committed, or the candidate may be released or rolled back. Actions at one control epoch therefore constrain the feasible actions and resource consumption at subsequent epochs. Moreover, multiple UE policies interact through finite satellite capacity and preparation resources. Our framework explicitly models this stateful protocol evolution and multi-user coupling under partial observability.

D. Agentic AI for Wireless and Satellite Networks

Agentic AI has recently emerged as a broader paradigm for autonomous network intelligence. Recent taxonomies associate Agentic AI with capabilities including contextual perception, persistent memory, reasoning, goal-directed behavior, autonomous action, and multi-agent coordination [21], [23]. The recent JSAC tutorial on future intelligent communications further describes Agentic AI as a closed-loop architecture that integrates perception, reasoning, planning, tools, and memory for continuous interaction with dynamic communication environments [22]. Advanced DRL is consequently receiving attention as an enabling technology for autonomous and adaptive wireless agents that must make sequential decisions [24].

Current communication-oriented Agentic-AI research follows several implementation paths. LLM-centric approaches exploit semantic reasoning and natural-language interaction. For example, Ni *et al.* employ an LLM as a spectrum-orchestration engine that combines spectrum observations with natural-language QoS descriptions to produce LEO resource-allocation strategies [25], and the Think–Do architecture separates high-level LLM reasoning from real-time execution for satellite resource orchestration [28]. Such approaches are particularly useful when a controller must interpret human intent or unstructured knowledge.

A complementary direction focuses on autonomous domain-specific agents whose perception and action spaces are directly tied to network control. AERS models individual satellite beams as autonomous agents that coordinate beam hopping, user scheduling, and power allocation under shared satellite constraints [26]. Xie *et al.* integrate multi-agent learning and task-oriented attention for distributed resource orchestration in satellite-assisted IoV [27]. More broadly, recent surveys identify AI/ML as an important enabler for adaptive satellite communication while emphasizing the computational constraints of on-board implementation [3].

TABLE I
MAIN NOTATIONS

Symbol	Description
$\mathcal{N}$, N; $\mathcal{L}$, L	Set and number of UEs; of satellites
$\mathcal{K}_i(t)$, K	Candidate set of UE i and its size
$\mathcal{A}$, $a_i(t)$	Action set and action of UE i
t, Δ, T	Epoch index, epoch duration, episode length
$\ell_i(t)$, $c_i(t)$	Serving and target satellite of UE i
$\gamma_{i\ell}$, $e_{i\ell}$, $q_{i\ell}$	SINR (dB), spectral efficiency, packet success prob.
$\epsilon_{i\ell}$, $d_{i\ell}$, $v_{i\ell}$	Elevation, slant range, visibility indicator
λ_i, D_i, Q_i	Arrival rate, demand, queue length
$x_{i\ell}$, $R_{i\ell}$, R_i	Requested resource fraction, leg rate, UE rate
κ_ℓ, ϖ	Capacity scale, reservation fraction per preparation
n_ℓ, m_ℓ, M	Reservations, new preparation requests, preparation slots
Φ_ℓ, U_ℓ	Requested pressure, aggregate intended demand
δ_i, u_i	Service deficit, outage indicator
α_i	Handover success probability
C_{def}, C_{ctrl}, C_{out}	Deficit, control, and outage costs
C_{thr}	Outage threshold
r, r_μ	Team reward, Lagrangian reward
$\mathbf{o}_i$, $\mathbf{b}_i$	Local observation, action mask
$\mathbf{g}_i$, $\boldsymbol{\zeta}_i$	Recurrent memory, experienced outcome
$\boldsymbol{\phi}_i$, $J_i(a)$	Reasoning feature, plan cost
y_ℓ, μ	Congestion price, reliability multiplier
π_θ, V_ψ	Actor, critic

Our work adopts this capability-oriented interpretation of Agentic AI. It does not place an LLM in the real-time mobility-control loop. Instead, domain-specific mechanisms instantiate the capabilities required by the considered problem: recurrent state provides temporal memory, deterministic ephemeris acts as a physics-grounded reasoning tool, congestion prices provide lightweight multi-agent coordination, and a dual controller supplies goal-oriented reliability regulation. The resulting architecture targets a protocol-control timescale at which predictable orbital dynamics and structured radio states can be exploited directly, without requiring unrestricted semantic reasoning or an LLM in the real-time control loop.

III. System Model

We consider the downlink of an NTN in which a set $\mathcal{L} = \{1, \ldots, L\}$ of LEO satellites serves a set $\mathcal{N} = \{1, \ldots, N\}$ of ground UEs. Time is slotted into control epochs $t \in \{0, 1, \ldots, T-1\}$ of duration Δ, and mobility decisions are taken once per epoch. Each UE is an autonomous decision-making agent that controls its own CHO and PDCP-duplication procedure, while all UEs share the finite radio and control-plane resources of the satellites. The main notations are summarized in Table I and the parameter values in Table II. Throughout the paper, calligraphic letters denote sets and bold letters denote vectors; a tilde marks a quantity observed by the agents through delayed and noisy telemetry, a hat marks a quantity predicted by the analytical planner of Section V, and an overbar marks an average.

Modeling scope and protocol abstraction. Our objective is to study the decision coupling among target preparation, PDCP duplication, and primary-cell switching rather than to reproduce the complete signalling stack. Accordingly, the CHO procedure is represented by a system-level state machine that abstracts the preparation, execution, completion, and cooldown phases of conditional mobility, while PDCP duplication is represented by the synchronized transmission of the same packet stream over the serving and the prepared target legs. This abstraction is inspired by the NR conditional-reconfiguration procedure of 3GPP TS 38.331 and the PDCP functionality of 3GPP TS 38.323 [14], [15]. It does not model individual RRC messages, RLC entities, HARQ processes, or radio-link-failure recovery. The radio scheduler is kept fixed throughout all experiments, so that the learned agents control only mobility preparation, duplication, and switching.

A. Constellation Geometry and Visibility

Satellite ℓ moves on a circular orbit of altitude h and inclination ι. Its Earth-centred position is

$$\mathbf{p}_\ell(t) = (E+h)\big[\cos\Omega_\ell(t),\, \sin\Omega_\ell(t)\cos\iota,\, \sin\Omega_\ell(t)\sin\iota\big]^{\mathsf{T}}, \tag{1}$$

where E is the Earth radius and $\Omega_\ell(t) = \Omega_\ell(0) \pm 2\pi t\Delta/T_{\mathrm{orb}}$ is the orbital phase for the orbital period T_{orb}. The initial phases $\Omega_\ell(0)$ are evenly spaced over $[-0.38, 0.38]$ rad, and the sign is positive for $\Omega_\ell(0) \le 0$ and negative otherwise. This construction represents a stylized local constellation slice with overlapping satellite passes rather than a full Walker constellation. It creates repeated rise–set transitions and simultaneous handover opportunities while keeping the mobility-control experiment computationally tractable. The UEs are fixed on the Earth surface, with latitudes and longitudes drawn uniformly within a ± 0.04 rad $\times$ ± 0.05 rad area. The elevation angle $\epsilon_{i\ell}(t)$ and the slant range $d_{i\ell}(t)$ follow from the positions of satellite ℓ and UE i, and satellite ℓ is visible to UE i if $v_{i\ell}(t) = \mathbb{1}\{\epsilon_{i\ell}(t) \ge \epsilon_{\min}\} = 1$.

B. Channel and Link Abstraction

The received power of the link (i, ℓ) in dBm is

$$P_{i\ell}(t) = P_{\mathrm{T}} + G_{\mathrm{T}} + G_{\mathrm{R}} - \mathrm{FSPL}\big(d_{i\ell}(t)\big) - A_{\mathrm{atm}} - A_{\mathrm{impl}} - A_{\mathrm{el}}\big(\epsilon_{i\ell}(t)\big) - \chi_{i\ell}(t) - B_{i\ell}(t) - Z_{i\ell}(t), \tag{2}$$

where P_{T} is the transmit power, G_{T} and G_{R} are the transmit and receive antenna gains, $\mathrm{FSPL}(d) = 20\log_{10} d + 20\log_{10} f - 147.55$ dB is the free-space path loss for the distance d in meters and the carrier frequency f in hertz, A_{atm} and A_{impl} are the atmospheric and implementation losses, and $A_{\mathrm{el}}(\epsilon) = 1.5/\max\{0.2, \sin\epsilon\}$ dB is the additional low-elevation loss. The shadowing $\chi_{i\ell}(t)$ is a first-order autoregressive (AR(1)) process with correlation coefficient ρ_χ and standard deviation σ_χ, $B_{i\ell}(t)$ is the extra loss caused by blockage events, and $Z_{i\ell}(t)$ is the loss of a hidden link-degradation regime (Section III-F). With the receiver noise power $P_{\mathrm{N}} = -174 + 10\log_{10} W + F$ dBm for the bandwidth W and the noise figure F, and a fixed aggregate interference floor P_{I}, the signal-to-interference-plus-noise ratio (SINR) of a visible link in dB is

$$\gamma_{i\ell}(t) = P_{i\ell}(t) - 10\log_{10}\big(10^{P_{\mathrm{N}}/10} + 10^{P_{\mathrm{I}}/10}\big). \tag{3}$$

Explicit co-channel interference dynamics are therefore outside the scope of the present model. The link is abstracted

by a capped Shannon efficiency and a smooth packet-success model,

$$e_{i\ell}(t) = \min\big\{e_{\max},\ 0.75\log_2\big(1+10^{\gamma_{i\ell}(t)/10}\big)\big\}, \tag{4}$$

$$q_{i\ell}(t) = \Big(1+\exp\big(-0.75\,(\gamma_{i\ell}(t)+2)\big)\Big)^{-1} v_{i\ell}(t), \tag{5}$$

where $e_{\max}$ is the spectral-efficiency cap and $q_{i\ell}$ is the per-packet success probability, which equals 0.5 at $-2\,\mathrm{dB}$.

C. Traffic, Scheduling, and Service

Packets of S bits arrive at UE i according to a Poisson process of rate $\lambda_i(t)$. The nominal rate of each UE is drawn uniformly within $\pm 35\%$ of the mean rate $\bar{\lambda}$, and it is multiplied by a burst factor during traffic-burst events. Packets wait in a finite queue of length $Q_i(t) \le Q_{\max}$ and are dropped when their age exceeds the deadline T_{dl}. A UE is active if $Q_i(t) > 0$, and its demand is

$$D_i(t) = \min\big\{\lambda_i(t),\ Q_i(t)S/\Delta\big\}. \tag{6}$$

Each UE has a serving satellite $\ell_i(t)$ and, during a mobility procedure, a target satellite $c_i(t)$. A UE requests a data leg on $\ell_i(t)$ and, when duplicating, a second leg on $c_i(t)$. The satellites run a fixed demand-proportional scheduler that is *not* controlled by the agents. For a leg (i,ℓ), the requested resource fraction is

$$x_{i\ell}(t) = \min\Big\{1,\ \frac{D_i(t)}{W\, e_{i\ell}(t)\,\kappa_\ell(t)}\Big\}, \tag{7}$$

where $\kappa_\ell(t)$ is the capacity scale of satellite ℓ; it equals the nominal value κ_0 except during capacity-drop events, in which it is reduced by a factor (and never falls below 0.05). We set $x_{i\ell}(t) = 0$ if UE i has no leg on satellite ℓ. Let $n_\ell(t)$ be the number of UEs holding a preparation reservation on satellite ℓ; each reservation blocks a fraction ϖ of the satellite resource, which leaves the fraction $1-\varpi n_\ell(t)$ for data. The scheduler grants every request when the total request fits into this fraction and scales all requests proportionally otherwise, which yields the leg rate

$$R_{i\ell}(t) = W e_{i\ell}(t)\kappa_\ell(t)\, x_{i\ell}(t) \min\Big\{1,\ \frac{1-\varpi n_\ell(t)}{\sum_{j\in\mathcal{N}} x_{j\ell}(t)}\Big\}. \tag{8}$$

The requested pressure of satellite ℓ is $\Phi_\ell(t) = \varpi n_\ell(t) + \sum_{i\in\mathcal{N}} x_{i\ell}(t)$. The scheduler treats all UEs equally and is intentionally kept outside the learned action space, so that performance differences among M1–M5 can be attributed to mobility preparation, duplication, reasoning, and coordination decisions rather than to radio-resource scheduling.

Delivery. Without duplication, UE i transmits at the rate $R_i = R_{i\ell_i}$ with the success probability $q_i = q_{i\ell_i}$. With PDCP duplication, the same packets are sent on both legs, so that $R_i = \min\{R_{i\ell_i}, R_{ic_i}\}$ and

$$q_i = (1-\varkappa)\big[1-(1-q_{i\ell_i})(1-q_{ic_i})\big] + \varkappa \max\{q_{i\ell_i}, q_{ic_i}\}, \tag{9}$$

where $\varkappa$ is a common-mode failure correlation. The number of delivered packets is $Y_i(t) \sim \mathrm{Bin}\big(\min\{\lfloor R_i(t)\Delta/S\rfloor, Q_i(t)\},\ q_i(t)\big)$.

Service metrics. The normalized service deficit and the outage indicator of an active UE are

$$\delta_i(t) = \frac{\big[D_i(t) - \min\{R_i(t), Q_i(t)S/\Delta\}\, q_i(t)\big]^+}{D_i(t)},$$
$$u_i(t) = \mathbb{1}\big\{Y_i(t)S < \beta D_i(t)\Delta\big\}, \tag{10}$$

where β is the minimum acceptable service ratio. Note that δ_i is a smooth expected-value quantity used for learning, whereas u_i is the realized, packet-level reliability event.

D. Conditional Handover and PDCP Duplication Protocol

Each UE runs the five-state mobility state machine of Fig. 1: SINGLE (S), PREPARED (P), DUPLICATING (D), SWITCHING (W), and COOLDOWN (C). The candidate set $\mathcal{K}_i(t)$ contains the K visible non-serving satellites with the highest *observed* SINR (Section III-E). The per-UE action set is

$$\mathcal{A} = \{\textsc{Stay},\ \textsc{Prep}_1, \ldots, \textsc{Prep}_K,\ \textsc{DupOn},$$
$$\textsc{DupOff},\ \textsc{Commit},\ \textsc{Release},\ \textsc{Rollback}\}, \tag{11}$$

which contains $K+6$ actions. A state-dependent hard mask $\mathbf{b}_i(t) \in \{0,1\}^{K+6}$ enforces protocol and physical validity:

- In S, $\textsc{Prep}_k$ requests CHO preparation towards the k-th candidate. If admitted, the UE enters P with the preparation timer T_{prep}.
- In P (after the timer expires), DUPON activates duplication (P→D), COMMIT executes the CHO (P→W), and RELEASE cancels the preparation (P→S).
- In D, DUPOFF returns to P, COMMIT executes the CHO, and ROLLBACK cancels both the duplication and the preparation.
- COMMIT is feasible only if the target is visible and the dwell condition $t - t_i \ge T_{\mathrm{dwell}}$ holds, where t_i is the epoch of the last completed switch of UE i.

If the target becomes invisible while the UE is in P or D, the preparation is lost and the UE returns to S. During the SWITCHING state, the source satellite remains the serving data leg until the switching outcome is resolved, while the target preparation reservation is retained. Hence, entering W does not immediately change the serving data path.

After T_{sw} epochs in W, the switch succeeds with the probability

$$\alpha_i(t) = \alpha_0\, q_{ic_i}(t)\, \exp\big(-[\Phi_{c_i}(t-1)-1]^+\big), \tag{12}$$

where α_0 is the base success probability and $\Phi_{c_i}(t-1)$ is the most recently realized requested pressure available when the switching outcome is resolved. The one-epoch offset follows from the causal control sequence: the switching outcome is resolved before the scheduler computes the requested pressure of the current epoch. The probability therefore decreases with poor target-link quality and recently observed target congestion.

On success, c_i becomes the new serving satellite and the UE enters C for T_{cd} epochs; otherwise it returns to S on the old serving satellite. A successful switch back to the previous serving satellite within T_{pp} epochs is counted as a ping-pong. Separate radio-link-failure reattachment is not modeled;

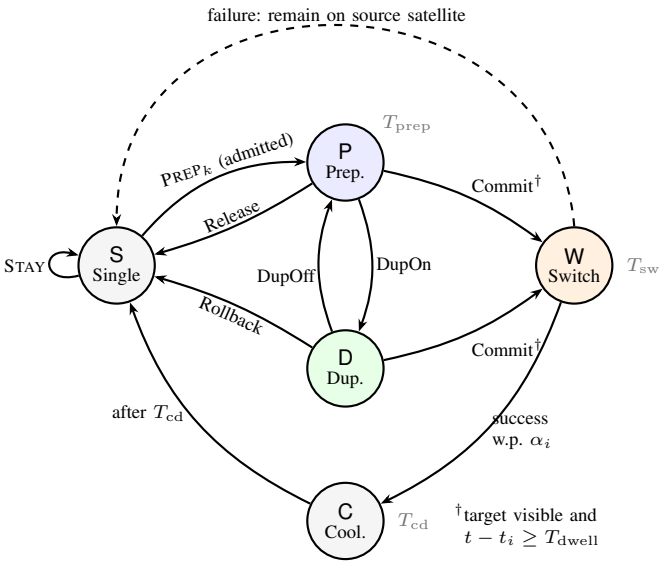


Fig. 1. Per-UE CHO and PDCP-duplication state machine. PREP$_k$ is subject to multi-UE admission; the timers T_{prep}, T_{sw}, and T_{cd} must elapse before leaving P, W, and C, respectively; a switch succeeds with the probability (12). Losing visibility of the target in P or D also returns the UE to S.

if the source data leg becomes unavailable before the switching outcome is resolved, the corresponding service may experience an outage.

Preparation admission. CHO preparation consumes two separate resources of the target satellite: the data-plane reservation ϖ and one of M control-plane preparation slots. All preparation requests of an epoch are collected first. Let $m_\ell(t)$ be the number of new requests towards satellite ℓ; if $m_\ell(t)$ exceeds $\min\{M - n_\ell(t),\ \lfloor(1 - \varpi n_\ell(t))/\varpi\rfloor\}$, the admitted requests are drawn uniformly at random and the others are rejected. Admission is therefore an explicit multi-UE contention.

E. Information Structure and Partial Observability

The agents do not observe the complete physical state. Current geometric link availability is assumed to be known for the serving-link visibility indicator and candidate presence, whereas radio-quality, elevation, and load information provided to the actor is delayed by T_{obs} epochs. In addition, the observed SINR and load are corrupted by temporally correlated noise. Specifically,

$$\tilde{\gamma}_{i\ell}(t) = \gamma_{i\ell}(t - T_{\mathrm{obs}}) + \nu_{i\ell}(t), \tag{13}$$
$$\tilde{e}_{i\ell}(t) = \min\Big\{e_{\max}, \max\big\{0,\, e_{i\ell}(t - T_{\mathrm{obs}})\, 2^{\nu_{i\ell}(t)/10}\big\}\Big\},$$
$$\tilde{\epsilon}_{i\ell}(t) = \epsilon_{i\ell}(t - T_{\mathrm{obs}}),$$
$$\tilde{\Phi}_{i\ell}(t) = \big[\Phi_\ell(t - 1 - T_{\mathrm{obs}}) + \omega_{i\ell}(t)\big]^+,$$

where the telemetry noises $\nu_{i\ell}(t)$ and $\omega_{i\ell}(t)$ are independent zero-mean AR(1) processes with correlation coefficients ρ_ν and ρ_ω and standard deviations σ_ν and σ_ω, respectively. Thus, elevation telemetry is delayed but is not subjected to an additional measurement-noise term in the present implementation. The local observation $\mathbf{o}_i(t) \in \mathbb{R}^{16+7K}$ contains (i) the queue length, head-of-line delay, arrival rate, reliability-class indicator, duplication flag, one-hot protocol state, and protocol timer; (ii) for the serving satellite: visibility, $\tilde{\gamma}$, $\tilde{e}$, $\tilde{\epsilon}$, and $\tilde{\Phi}$; and (iii) for each candidate: presence, a prepared-target flag, $\tilde{\gamma}$, $\tilde{e}$, $\tilde{\epsilon}$, $\tilde{\Phi}$, and the observed SINR trend $\tilde{\gamma}(t) - \tilde{\gamma}(t-1)$. Agents do *not* observe the remaining visibility time, the capacity scales, the event realizations, the latent link regimes, or the states and actions of other UEs. The candidate list is ranked by observed current SINR only, and the action mask does not check whether a candidate will remain visible long enough to complete a handover: anticipating the satellite geometry is left to the decision layer. The per-UE reliability-class indicator is fixed to 0.99 in the present experiments and should not be confused with the system-level outage threshold $C_{\mathrm{thr}} = 0.10$ introduced in Section IV.

Per-epoch control sequence. At decision epoch t, UE i forms its local observation from delayed telemetry, its current protocol state, and its locally experienced history; in M4 and M5 it additionally receives the satellite congestion prices computed at the end of the preceding epoch. Each UE then selects one masked mobility action independently. The environment first resolves ongoing protocol timers and switching outcomes, re-validates the selected actions (actions that have become infeasible are replaced by STAY), and jointly admits the simultaneous preparation requests. New packets arrive, the fixed scheduler allocates the remaining satellite resources to the serving and duplicated data legs, and packet delivery, service deficit, and outage are realized. The resulting local outcomes are fed to the recurrent memory for the next decision. Finally, the hidden regimes and disturbance events advance, the links and telemetry are refreshed, and the satellite coordinator updates its congestion price from the aggregate intended demand, so that the new price affects only the next decentralized decision. Coordination is therefore causal and one-epoch delayed rather than based on instantaneous access to the actions of other UEs.

F. Non-Stationarity: Hidden Regimes and Disturbance Events

Hidden link regimes. Every UE–satellite path (i, ℓ) follows a hidden semi-Markov regime GOOD $\rightarrow$ DEGRADED $\rightarrow$ RECOVERING $\rightarrow$ GOOD. A path enters DEGRADED with a small probability per epoch and with a larger probability at the start of an episode (Table II), stays there for a duration drawn uniformly between 8 and 24 epochs with an extra loss $Z_{i\ell}$ drawn uniformly between 7 and 15 dB, and then recovers over 4 to 10 epochs, during which $Z_{i\ell}$ decays linearly to zero. These regimes persist over many epochs and are observable only through their delayed, noisy effect on $\tilde{\gamma}$, which is what makes memory useful.

Disturbance events. Each training and nominal-evaluation episode contains five events whose start, duration, severity, affected satellite, and affected UE subset are sampled from the ranges in Table II: two blockages (extra loss $B_{i\ell}$ on the selected UE–satellite links), two capacity drops (reduced capacity scale κ_ℓ of one satellite), and one traffic burst (increased arrival rate λ_i of a subset of UEs). For the *compound-stress* evaluation, the randomized training-event schedule is

TABLE II
SIMULATION AND LEARNING PARAMETERS

Network and protocol	
Satellites / UEs / candidates	$L = 6$, $N = 12$, $K = 3$
Altitude, inclination, period	$h = 600\,\text{km}$, $\iota = 53^\circ$, $T_{\text{orb}} = 5800\,\text{s}$
Epoch, episode, min. elev.	$\Delta = 1\,\text{s}$, $T = 420$, $\epsilon_{\min} = 10^\circ$
Carrier, bandwidth	$f = 2\,\text{GHz}$, $W = 4\,\text{MHz}$
Tx power, Tx/Rx gain	$P_{\text{T}} = 46\,\text{dBm}$, $G_{\text{T}} = 30\,\text{dBi}$, $G_{\text{R}} = 0\,\text{dBi}$
Noise figure, interference	$F = 7\,\text{dB}$, $P_{\text{I}} = -108\,\text{dBm}$
Losses (atm./impl.)	$A_{\text{atm}} = 0.5\,\text{dB}$, $A_{\text{impl}} = 3\,\text{dB}$
Shadowing	$\sigma_\chi = 2\,\text{dB}$, $\rho_\chi = 0.95$
SE cap, capacity scale	$e_{\max} = 6$, $\kappa_0 = 0.7$
Packet size, deadline, queue	$S = 12\,\text{kbit}$, $T_{\text{dl}} = 2\,\text{s}$, $Q_{\max} = 400$
Mean rate, spread	$\bar{\lambda} = 2\,\text{Mb/s}$, $\pm 35\%$
Min. service ratio, reliab. class	$\beta = 0.8, 0.99$ (all UEs)
$T_{\text{prep}}, T_{\text{sw}}, T_{\text{cd}}, T_{\text{dwell}}, T_{\text{pp}}$	$1, 2, 5, 15, 20$ epochs
Prep. reservation, control slots	$\varpi = 0.01$, $M = 2$
HO success, dup. corr.	$\alpha_0 = 0.995$, $\varkappa = 0.15$
Partial observability and non-stationarity	
Telemetry lag	$T_{\text{obs}} = 2$ epochs
Telemetry noise	$\sigma_\nu = 1.5\,\text{dB}$, $\sigma_\omega = 0.08$, $\rho_\nu = \rho_\omega = 0.8$
Regime entry probability	0.06 (episode start), 0.006 (per epoch)
Blockage (2 events)	8–15 dB, 25–55 epochs, 30–65% of UEs
Capacity drop (2 events)	factor 0.45–0.80, 25–70 epochs
Traffic burst (1 event)	factor 1.35–1.85, 35–70 epochs
Objective and agentic modules	
Cost weights	$w = 0.25$, $w_{\text{prep}} = 0.2$, $w_{\text{HO}} = 0.05$
Outage threshold	$C_{\text{thr}} = 0.10$
Dual step size, bound	$\eta_\mu = 0.005$, $\mu_{\max} = 10$
Price step size, bound	$\eta_y = 0.05$, $y_{\max} = 5$
Planning horizon, discount	$\tau_{\max} = 6$, $\xi_{\text{p}} = 0.95$
Learning (CTDE-MAPPO)	
Actor / critic MLP	$[128, 128]$ / $[256, 256]$
GRU hidden size, TBPTT length	128, 16
Rollout, iterations	$T_{\text{roll}} = 128$, $I = 2000$
PPO epochs, minibatch	5, 128
Discount, GAE factor, clip, KL	$\xi = 0.99, 0.95, 0.2, 0.03$
LR actor/critic	$3{\cdot}10^{-4} \to 3{\cdot}10^{-5}$ / $10^{-3} \to 10^{-4}$
Entropy / value coef., grad. norm	0.01 / 0.5, 0.5

replaced by a deterministic overlapping disturbance pattern: a traffic burst with the factor 2.5 on 50% of the UEs during $t \in [60, 120)$, a capacity drop of satellite 2 to 45% of its nominal capacity scale during $t \in [80, 130)$, and an 18-dB blockage affecting all links of 25% of the UEs during $t \in [100, 125)$. The traffic-burst and blockage severities lie outside their training ranges, and the blockage affects all satellites of a UE instead of a single one, whereas the capacity-drop magnitude equals the most severe value of its training range. More importantly, the three disturbances overlap in time, creating a compound operating condition not encountered during training.

IV. Problem Formulation

A. Per-Epoch Costs

Let $\mathcal{N}(t) \subseteq \mathcal{N}$ be the set of UEs that are active in epoch t. We define the service, control, and reliability costs of epoch t as

$$C_{\text{def}}(t) = \frac{1}{|\mathcal{N}(t)|} \sum_{i \in \mathcal{N}(t)} \delta_i(t), \tag{14}$$

$$C_{\text{prep}}(t) = \frac{\varpi}{L} \sum_{\ell \in \mathcal{L}} n_\ell(t), \tag{15}$$

$$C_{\text{HO}}(t) = \frac{1}{|\mathcal{N}(t)|} \sum_{i \in \mathcal{N}(t)} \mathbb{1}\{a_i(t) = \textsc{Commit}\}, \tag{16}$$

$$C_{\text{ctrl}}(t) = C_{\text{dup}}(t) + w_{\text{prep}} C_{\text{prep}}(t) + w_{\text{HO}} C_{\text{HO}}(t), \tag{17}$$

$$C_{\text{out}}(t) = \frac{1}{|\mathcal{N}(t)|} \sum_{i \in \mathcal{N}(t)} u_i(t), \tag{18}$$

where the duplication cost $C_{\text{dup}}(t)$ is the fraction of the bits scheduled for transmission in epoch t that are duplicate copies sent over secondary legs. Hence, C_{dup} measures transmission redundancy rather than successfully delivered traffic. The control cost C_{ctrl} is a resource- and mobility-control-cost proxy comprising duplicated transmissions, target preparation reservations, and handover executions with the weights w_{prep} and w_{HO}. It does not explicitly charge the number of preparation signalling messages; this distinction is revisited in Section VIII.

B. Constrained Decentralized POMDP

The problem is a cooperative, constrained decentralized partially observable Markov decision process (Dec-POMDP) with the agent set $\mathcal{N}$, the state space $\mathcal{S}$, the action set $\mathcal{A}$ of every agent, the transition kernel $\mathcal{P}$, the team reward r, and the discount factor ξ. The state $s(t) \in \mathcal{S}$ comprises the satellite geometry, channel and shadowing states, hidden regimes, event realizations, queues, and protocol states of all UEs; $\mathcal{P}$ is given by Section III; and each agent receives the local observation $\mathbf{o}_i(t)$ of Section III-E together with its action mask. All agents share the team reward

$$r(t) = -C_{\text{def}}(t) - w\, C_{\text{ctrl}}(t), \tag{19}$$

where w weights the control cost against the service deficit. Reliability is imposed as a constraint on the long-run outage cost rather than as a further reward weight:

$$\max_{\boldsymbol{\pi}} \ \mathbb{E}_{\boldsymbol{\pi}}\Big[\sum_t \xi^t r(t)\Big] \ \text{ s.t. } \ \mathbb{E}_{\boldsymbol{\pi}}\big[C_{\text{out}}(t)\big] \le C_{\text{thr}}, \tag{20}$$

where C_{thr} is the outage threshold and $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_N)$ are decentralized policies that map the local observation history of each UE to its action. The per-satellite capacity constraints are enforced by the scheduler and the admission rule and need not be learned. Problem (20) is solved through its Lagrangian relaxation

$$\max_{\boldsymbol{\pi}} \min_{\mu \ge 0} \ \mathbb{E}_{\boldsymbol{\pi}}\Big[\sum_t \xi^t r_\mu(t)\Big], \tag{21}$$
$$r_\mu(t) = r(t) - \mu\big(C_{\text{out}}(t) - C_{\text{thr}}\big),$$

where μ is the reliability multiplier.

Why the problem is hard. Three features make (20) difficult for conventional CHO rules and for memoryless learning: (i) *partial observability* with delayed, noisy telemetry and

TABLE III
CAPABILITY-STRUCTURED MODEL FAMILY

Model	Capability added	Actor input $\mathbf{z}_i$	Training reward
M1	Perception (core)	$\mathbf{o}_i$	r
M2	+ Memory	$[\mathbf{o}_i, \mathbf{g}_i]$	r
M3	+ Reasoning	$[\mathbf{o}_i, \mathbf{g}_i, \boldsymbol{\phi}_i]$	r
M4	+ Coordination	$[\mathbf{o}_i, \mathbf{g}_i, \boldsymbol{\phi}_i, \mathbf{y}_i/y_{\max}]$	r
M5	+ Goal regulation	$[\mathbf{o}_i, \mathbf{g}_i, \boldsymbol{\phi}_i, \mathbf{y}_i/y_{\max}, \mu/\mu_{\max}]$	r_μ

hidden, persistent link regimes; (ii) *delayed consequences*: a preparation blocks resources and control slots before any benefit appears, the dwell guard restricts future switches, and a target chosen now may set before the switch completes; and (iii) *coupling*: UEs compete for preparation slots, reservations, and data capacity, and congestion lowers the handover success probability (12).

V. PROPOSED CAPABILITY-STRUCTURED AGENTIC FRAMEWORK

We operationalize agentic control through a structured set of capabilities rather than treating agentic AI as synonymous with an LLM-based controller. The architecture combines a learned decentralized policy with lightweight domain-specific mechanisms that exploit the structure of LEO mobility, following the four capabilities of agentic AI. *Perception* is provided by masked, telemetry-based local observation and per-UE autonomous action selection; *memory* by a recurrent state that infers persistent hidden conditions from delayed feedback; *reasoning* by a protocol-aware ephemeris lookahead that anticipates the delayed consequences of preparation and switching; and *coordination* by a low-dimensional satellite congestion price that summarizes shared-resource pressure without exchanging UE actions. Finally, a training-time reliability multiplier provides explicit *goal regulation*.

These capabilities are introduced incrementally to form the M1–M5 model family in Table III; Fig. 2 shows the resulting agent. The resulting ladder is an architectural decomposition used for capability-structured evaluation, not an assumption that performance must improve monotonically from M1 to M5. Each variant is trained independently in the same environment with the same base reward, except for the reliability Lagrangian used by M5. No language model is required, and online execution remains numerical and lightweight: actions are produced locally by each UE, while satellites broadcast only the scalar congestion prices required by M4 and M5.

A. Perception and Core Decision Making (M1)

All UEs share one stochastic actor π_θ with parameters θ and are trained with centralized training and decentralized execution (CTDE) using multi-agent proximal policy optimization (MAPPO) [29]. The actor maps its input $\mathbf{z}_i(t)$ to one logit per action through a neural network NN_θ, and invalid actions are removed by a masked softmax,

$$\pi_\theta(a \mid \mathbf{z}_i, \mathbf{b}_i) = \frac{b_{i,a} \exp\big([\mathrm{NN}_\theta(\mathbf{z}_i)]_a\big)}{\mathbf{b}_i^{\mathsf{T}} \exp\big(\mathrm{NN}_\theta(\mathbf{z}_i)\big)}, \tag{22}$$

where $b_{i,a}$ is the entry of $\mathbf{b}_i$ for action a and the exponential in the denominator is applied elementwise. Perception in M1 is the masked, telemetry-based observation $\mathbf{z}_i = \mathbf{o}_i$. The centralized critic V_ψ with parameters ψ is used only during training. Its input $\mathbf{s}(t)$ concatenates the observations of all UEs with the true per-satellite utilization and pressure Φ_ℓ, and, when available, the satellite prices and the reliability multiplier.

B. Memory: Inferring Hidden Regimes (M2)

Because link regimes are hidden and telemetry is delayed, the current observation is not a sufficient statistic. M2 equips every agent with a gated recurrent unit (GRU) that summarizes its interaction history in the recurrent memory

$$\mathbf{g}_i(t) = \mathrm{GRU}_\theta\big(\mathbf{g}_i(t{-}1),\, [\mathbf{o}_i(t),\, \mathrm{onehot}(a_i(t{-}1)),\, \boldsymbol{\zeta}_i(t{-}1)]\big), \tag{23}$$

where $\boldsymbol{\zeta}_i \in \mathbb{R}^6$ is the agent's own *experienced outcome*: the last handover result ($+1$ success, -1 failure, 0 none), its outage indicator u_i, its deficit δ_i, the duplication gain (the part of its expected goodput contributed by the secondary leg, normalized by D_i), its resource-usage indicator, and the experienced congestion $\Phi/(1+\Phi)$ of its serving or target satellite. The actor input becomes $\mathbf{z}_i = [\mathbf{o}_i, \mathbf{g}_i]$. Memory thus supports long-horizon adaptation to non-stationary link conditions using only locally available feedback. The GRU is trained end-to-end with truncated backpropagation through time.

C. Reasoning: Protocol-Aware Ephemeris Lookahead (M3)

The mask does not prevent choosing a target that is about to set, and the benefit of a preparation appears only several epochs later. M3 therefore adds an analytical planner that exploits the one piece of information that is deterministic in LEO systems: the satellite ephemeris. For every feasible first action a, the planner evaluates a small library $\mathcal{Q}_i(a)$ of protocol-consistent plans, i.e., action sequences $\mathbf{a} = (a(0), \ldots, a(\tau_{\max}-1))$ of length $\tau_{\max}$ that start with a (e.g., PREP$_k$–wait–COMMIT; PREP$_k$–wait–DUPON–COMMIT; STAY) and are padded with STAY, by rolling the UE's own protocol state machine forward. For tractability, this hypothetical rollout propagates protocol timers deterministically. Once a hypothetical COMMIT satisfies the dwell and predicted-visibility conditions, completion of the corresponding switch is treated as successful in the lookahead. Consequently, the stochastic handover-success model in (12) is realized by the environment but is not explicitly sampled or averaged inside the analytical planner.

The SINR of link (i,ℓ) at the lookahead step $\tau \in \{0, \ldots, \tau_{\max}-1\}$ is predicted from the *observed* SINR and the ephemeris-predicted range,

$$\hat{\gamma}_{i\ell}(t+\tau) = \tilde{\gamma}_{i\ell}(t) - 20\log_{10}\frac{d_{i\ell}(t+\tau)}{d_{i\ell}(t)}, \tag{24}$$

and the link is treated as unavailable once $\epsilon_{i\ell}(t+\tau) < \epsilon_{\min}$. The predicted leg rate uses the nominal capacity scale κ_0 scaled by the observed load $\tilde{\Phi}_{i\ell}$, from which the planner

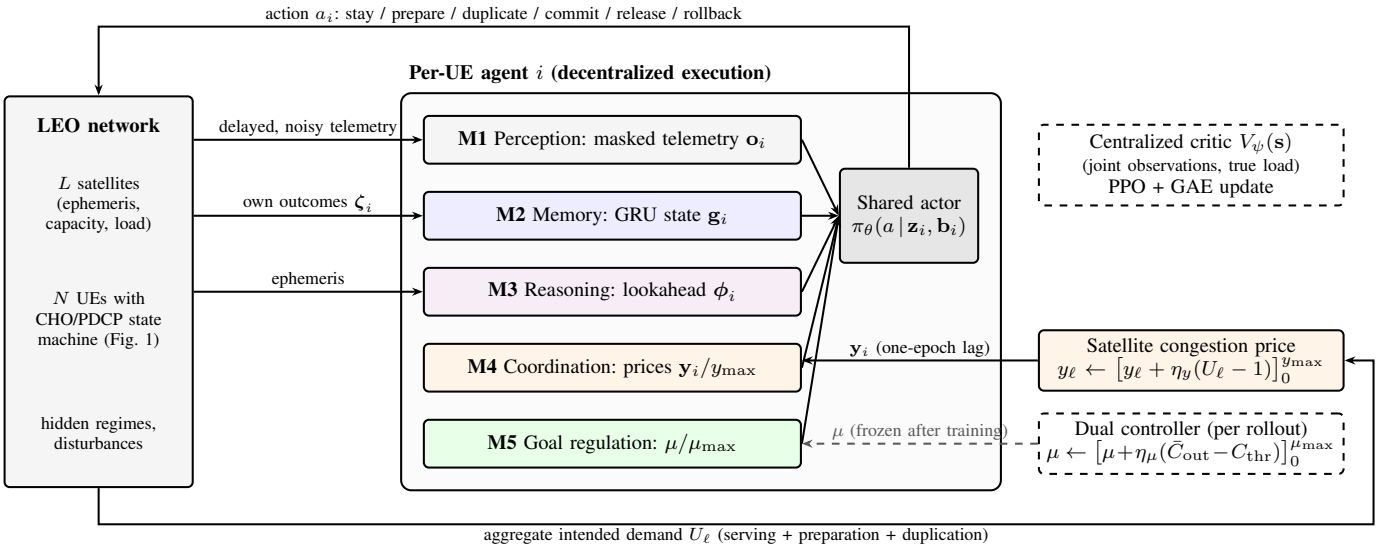


Fig. 2. Architecture of the capability-structured agent. Each UE runs the same actor on the concatenation $\mathbf{z}_i$ of the capability outputs available in its variant (M1: $\mathbf{o}_i$; M2 adds $\mathbf{g}_i$; M3 adds $\boldsymbol{\phi}_i$; M4 adds $\mathbf{y}_i/y_{\max}$; M5 adds $\mu/\mu_{\max}$). Solid: execution; dashed: training only. Satellites broadcast only the scalar prices.

obtains the predicted deficit $\hat{\delta}_i(t+\tau \mid \mathbf{a})$ under plan $\mathbf{a}$, including the duplication diversity of (9). The cost of a first action is the best discounted plan cost,

$$J_i(a) = \min_{\mathbf{a}\in\mathcal{Q}_i(a)} \sum_{\tau=0}^{\tau_{\max}-1} \xi_{\mathrm{p}}^{\tau} \Big(\hat{\delta}_i(t+\tau \mid \mathbf{a}) + \mathbf{y}(t)^{\mathsf{T}} \big[\hat{\mathbf{x}}_i(t+\tau \mid \mathbf{a}) - \hat{\mathbf{x}}_i(t+\tau \mid \mathbf{a}_0) \big]^{+} \Big), \quad (25)$$

where ξ_{p} is the planning discount, $\mathbf{a}_0$ is the plan that stays throughout, and the second term is present only in M4 and M5. Here, $\mathbf{y}(t) = [y_1(t), \ldots, y_L(t)]^{\mathsf{T}}$ denotes the *raw* satellite-price vector maintained by the coordinator. Only entries corresponding to the UE's serving and hypothetical target satellites contribute to the inner product.

The vector $\hat{\mathbf{x}}_i$ represents the planner's predicted per-satellite resource footprint of UE i. In the present implementation it contains the predicted serving-leg resource fraction, the secondary-leg resource fraction while duplicating, and the data-plane reservation ϖ while a target reservation is active. The positive-part difference with the footprint of $\mathbf{a}_0$ therefore prices only the incremental resource footprint relative to remaining with the current serving satellite.

Importantly, $\hat{\mathbf{x}}_i$ is not identical to the aggregate intended demand U_ℓ used by the satellite coordinator (Section V-D). The control-plane preparation-contention term m_ℓ/M is included in U_ℓ when the satellite price is updated, but is not explicitly inserted into the single-UE hypothetical footprint. Its effect therefore reaches the reasoner indirectly through the broadcast congestion price.

The reasoner deliberately ranks plans primarily by predicted service continuity rather than reproducing the complete MAPPO training reward. In M3, the second term of (25) is absent, so resource and mobility-control costs are learned through the team reward rather than hard-coded into the planner. M4 and M5 add the marginal raw-price term, allowing the same geometry-aware lookahead to account for recently observed shared resource pressure.

Let $J_{\min}$ and $J_{\max}$ be the minimum and maximum of $J_i(a)$ over the feasible actions of UE i. The reasoning feature of a feasible action is

$$\phi_{i,a} = \begin{cases} \dfrac{J_i(a) - J_{\min}}{J_{\max} - J_{\min}}, & J_{\max} - J_{\min} > 10^{-8}, \\ 0, & \text{otherwise}, \end{cases} \quad (26)$$

and $\phi_{i,a} = 1$ for masked actions. The vector $\boldsymbol{\phi}_i \in [0,1]^{K+6}$ is appended to the actor input: $\mathbf{z}_i = [\mathbf{o}_i, \mathbf{g}_i, \boldsymbol{\phi}_i]$. The planner is *advisory*: the actor learns how much to trust it. It has no access to future events, hidden regimes, true capacity scales, or the states of other UEs.

D. Coordination: Congestion Prices (M4)

UEs compete for preparation slots, reservations, and data capacity, but exchanging intentions between UEs is impractical. M4 coordinates them through a lightweight price signal maintained by each satellite. Let $\mathcal{D}(t)$ be the set of UEs that duplicate in epoch t. Satellite ℓ measures its aggregate intended demand

$$U_\ell(t) = \sum_{i:\, \ell_i(t)=\ell} x_{i\ell}(t) + \frac{m_\ell(t)}{M} + \sum_{i\in\mathcal{D}(t):\, c_i(t)=\ell} x_{i\ell}(t), \quad (27)$$

i.e., the serving demand, the new preparation requests normalized by the number of control slots, and the duplication demand, and updates its price y_ℓ by projected subgradient ascent with the step size η_y,

$$y_\ell(t+1) = \Big[y_\ell(t) + \eta_y \big(U_\ell(t) - 1 \big) \Big]_0^{y_{\max}}. \quad (28)$$

Prices rise while intended demand exceeds the normalized resource level and decay otherwise; they are reset at the start of each episode and are not learned. The price produced at the end of epoch t is used only for the next decentralized decision, so coordination is causal and one-epoch delayed.

The actor of UE i receives the local price vector $\mathbf{y}_i(t) = [y_\ell(t)]_{\ell \in \{\ell_i(t)\} \cup \mathcal{K}_i(t)}$ of its serving satellite and its K candidates, normalized as $\mathbf{y}_i(t)/y_{\max}$ to provide a bounded neural-network input. In contrast, the analytical lookahead in (25) uses the raw prices, which directly weight the planner's incremental resource demand.

The coordinator never allocates actions: decisions remain decentralized, and the signalling required for coordination is one scalar price per satellite and control epoch. The price should therefore be interpreted as compressed coordination context rather than a centralized allocation command. It reveals neither individual UE states, actions, queues, nor future events, and conveys only the recent aggregate resource pressure seen by each satellite.

E. Goal Regulation: Primal–Dual Reliability Control (M5)

M5 addresses the Lagrangian problem (21). The dual update uses the realized outage fraction over all active-UE opportunities in one rollout. For the set $\mathcal{T}$ of the T_{roll} epochs of a rollout, the implementation computes the rollout-average outage cost

$$\bar{C}_{\mathrm{out}} = \frac{\sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{N}(t)} u_i(t)}{\sum_{t \in \mathcal{T}} |\mathcal{N}(t)|}. \tag{29}$$

After every rollout, the multiplier is updated by the projected step

$$\mu \leftarrow \Big[\mu + \eta_\mu \big(\bar{C}_{\mathrm{out}} - C_{\mathrm{thr}}\big)\Big]_0^{\mu_{\max}}, \tag{30}$$

and the policy is trained on r_μ. The multiplier is held fixed within each rollout, updated between policy iterations, and frozen after training. In our implementation, μ does not modify the ranking of the analytical lookahead; reliability enters the learned decision through the Lagrangian reward, the recurrently observed outcomes, and the normalized feature $\mu/\mu_{\max}$ supplied to the actor and the critic. Consequently, $\mu = 0$ makes the *instantaneous training reward* identical to the M4 base reward, but a trained M5 network remains a separately optimized policy and is not identical to M4.

F. Training Procedure and Complexity

Algorithm 1 summarizes training. The shared actor and the centralized critic are optimized with the clipped PPO objective [30] and generalized advantage estimation [31] on the team reward (r or r_μ), with early stopping at a target KL divergence and linearly annealed learning rates.

Execution complexity. In each control epoch, a UE performs one recurrent-state update and one actor forward pass. For M3–M5, the analytical reasoner evaluates at most $\max\{2K + 1, 4\}$ protocol-consistent plans, each over $\tau_{\max}$ epochs. Its per-agent planning complexity is therefore $\mathcal{O}(K\tau_{\max})$ and does not grow with the total number of UEs. Across all N agents, the aggregate reasoning work is $\mathcal{O}(NK\tau_{\max})$, but the N per-UE computations are mutually independent and can be executed in parallel. Coordination requires only one congestion price per satellite and epoch, and the M5 reliability multiplier introduces no execution-time inter-UE signalling. Measured on a single CPU thread of a desktop processor with an unoptimized Python/NumPy implementation, the mean actor inference time per epoch for all $N = 12$ UEs is 0.24–0.44 ms, and the analytical reasoner adds 19.6–21.3 ms. The total mean decision latency of M3–M5 is thus at most 21.7 ms, i.e., about 1.8 ms per UE, which is more than an order of magnitude below the $\Delta = 1$ s control epoch.

Algorithm 1 Training of the capability-structured agents

```
Initialize θ (actor, GRU), ψ (critic), μ ← 0
for iteration = 1, ..., I do
    for each epoch t ∈ T of the rollout do
        Each UE forms o_i(t) and b_i(t)                ▷ perception
        g_i(t) ← GRU_θ(·)                              ▷ M2+
        φ_i(t) ← lookahead (25) with y(t)              ▷ M3+
        a_i(t) ~ π_θ(· | z_i(t), b_i(t)) for all i in parallel
        Environment step; observe r(t), C_out(t), ζ_i(t)
        Satellites update prices by (28)               ▷ M4+
    end for
    Compute r_μ with the current μ                     ▷ M5
    Update θ, ψ with PPO, GAE, and TBPTT
    Update μ by (30)                                   ▷ M5
end for
```

VI. Simulation Setup

A. Training

All variants use the environment and parameters of Table II. Each variant is trained for $I = 2000$ iterations with five independent training seeds $\{7, 8, 9, 10, 11\}$; the seed determines both the network initialization and the training-environment realization. The five M1–M5 variants therefore comprise 25 independently trained policies. Training episodes contain the randomized disturbance schedule of Section III-F; the compound-stress pattern is never used for training. To isolate goal regulation from the other capabilities, we additionally train an ablation variant, denoted M1D, which is the memoryless M1 trained with the Lagrangian reward r_μ and the same dual controller as M5, using the same five seeds. The actor input of M1D is unchanged, and μ enters only its critic. M1D is not part of the capability ladder.

B. Evaluation Protocol

Every trained policy is evaluated on five held-out evaluation seeds with 40 episodes each, under two conditions: *nominal* (randomized events drawn as in training) and *compound stress* (Section III-F). The episode streams of different evaluation seeds are disjoint, and all methods are evaluated on identical episode streams, which yields paired comparisons. Actions are sampled from the stochastic policies; congestion prices evolve online and are reset at the start of each episode, and the multiplier of M5 is frozen at its trained value. For a learned variant, each training seed is first averaged over the five evaluation seeds, and the mean and the Student-t 95% confidence interval (CI) are computed across the five training seeds, so the CI reflects training variability. Differences between learned variants are tested with Welch's t-test on the five per-seed means, and "significant" refers to the 5% level. Rule-based baselines have no training seed; their CI

is computed across the evaluation seeds and mainly reflects the difficulty of the individual scenarios. Besides the realized outage ratio, we report the service deficit, the number of handover executions per epoch (summed over the N UEs), the ping-pong ratio (ping-pongs per successful handover), the resource cost $C_{\mathrm{ctrl}} = C_{\mathrm{dup}} + w_{\mathrm{prep}} C_{\mathrm{prep}} + w_{\mathrm{HO}} C_{\mathrm{HO}}$, and the *common reward* $r = -C_{\mathrm{def}} - w C_{\mathrm{ctrl}}$ of (19). The common reward is used for all methods, including M5, whose training reward additionally contains the Lagrangian term. Two within-policy ablations are evaluated on the same episodes: forcing the congestion price to zero (M4 and M5), and replacing the reasoning feature by an uninformative one (M3–M5), i.e., $\phi_{i,a} = 0$ for every feasible action and $\phi_{i,a} = 1$ for every masked action, which is the value the actor observes during training whenever all plan costs are tied.

C. Baselines

All baselines observe the same delayed and noisy telemetry and are subject to the same action masks and admission rules as the learned agents.

- *CHO (A3/TTT)*: a 3GPP-inspired conditional handover. The best observed candidate is prepared when it exceeds the serving satellite by 1 dB; the prepared target is executed when it exceeds the serving satellite by an execution offset for a number of consecutive epochs given by the time-to-trigger (TTT), and it is released after it falls below the serving satellite by 1 dB for three epochs. The execution offset (2, 3, or 4 dB) and the TTT (1, 2, or 3 epochs) were tuned for the common reward on three validation seeds whose episodes are disjoint from all evaluation episodes, and then frozen at 2 dB and one epoch. The selected offset is the smallest value of the grid, and a TTT of one epoch is the smallest feasible value at $\Delta = 1$ s.
- *CHO (A3/TTT)+Dup*: the tuned CHO rule, additionally activating duplication while a target is prepared and the serving SINR is below a threshold tuned in the same way (6 dB, selected from $\{2, 4, 6\}$ dB).
- *Max-SINR*: prepares the best observed candidate when it exceeds the serving satellite by 1 dB and executes as soon as allowed, without duplication.
- *Always-Dup*: always prepares the best observed candidate, activates duplication, and executes.
- *Threshold-Dup*: prepares the best candidate when the serving SINR falls below 4 dB, activates duplication, and executes when the target exceeds the serving satellite by 1 dB.

Learning-based handover schemes from the literature [16]–[18], [20] select a serving satellite or trigger a handover, but they do not expose preparation, duplication, release, and roll-back actions and therefore cannot be applied to the considered action space without redesign. As the learned reference, we therefore use M1, a CTDE-MAPPO policy [29] trained on the same observation space, action space, and reward as the proposed agents but without any of the agentic extensions; it is denoted MAPPO (M1) in the comparisons.

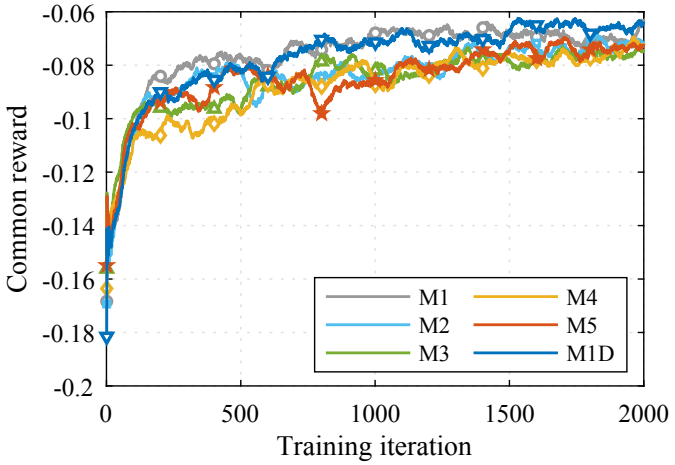


Fig. 3. Training convergence in terms of the common reward $r = -C_{\mathrm{def}} - wC_{\mathrm{ctrl}}$ (moving average over 50 iterations, mean over five training seeds). M5 and M1D are trained on the Lagrangian reward r_μ but are shown with the common reward for comparability.

VII. Results

A. Training Convergence

Fig. 3 shows the common reward during training. All variants improve from approximately -0.13 to -0.07 within the first few hundred iterations and then refine slowly under the annealed learning rates. Averaged over the last 100 iterations, M1 attains the highest training common reward (-0.067), M2, M4, and M5 reach about -0.072, and M3 reaches -0.075; this ordering mirrors the common reward on the held-out scenarios (Table IV). M5 is not trained on the common reward but on the Lagrangian reward r_μ, whose logged value additionally contains the reliability penalty $\mu(C_{\mathrm{out}} - C_{\mathrm{thr}})$ and is therefore not comparable across variants; in terms of the common reward, M5 is on par with M2 and M4 while being markedly more efficient in mobility and resource usage. Because each training seed also defines a different training-environment realization, the curves are compared qualitatively, and the variants are compared quantitatively only on the common held-out evaluation scenarios.

B. Comparison with Baselines

Table IV and Fig. 4 compare M5 with all five rule-based schemes and with the MAPPO policy M1. Both learned policies attain a far higher reliability than every rule-based scheme. In the nominal condition, the learned variants attain outage ratios of 0.131–0.149, whereas the rules attain 0.322–0.429. Relative to the five rules, M5 lowers the outage ratio by 58–68% and the service deficit by 49–64%, raises the packet delivery ratio from 80.3–85.6% to 92.6%, and at least halves the deadline-violation rate (2.2% versus 4.3–7.5%); under compound stress, the outage reduction is 44–56% (46% relative to CHO). M5 also has a 79–91% lower ping-pong ratio and 67–79% fewer failed handovers than CHO, CHO+Dup, Max-SINR, and Always-Dup. Threshold-Dup attains lower mobility counts only because it rarely hands over (0.09 executions per epoch), which leaves UEs on degrading links

TABLE IV

PERFORMANCE UNDER NOMINAL AND COMPOUND-STRESS CONDITIONS (MEAN $\pm$ 95% CI HALF-WIDTH). LEARNED MODELS: CI ACROSS 5 INDEPENDENTLY TRAINED POLICIES, EACH EVALUATED ON 5 EVALUATION SEEDS $\times$ 40 EPISODES. HEURISTICS: CI ACROSS THE 5 EVALUATION SEEDS. HO EXECUTIONS ARE COUNTED PER CONTROL EPOCH OVER ALL $N = 12$ UES; THE PING-PONG RATIO IS RELATIVE TO SUCCESSFUL HANDOVERS; COMMON REWARD $r = -C_{\mathrm{def}} - wC_{\mathrm{ctrl}}$. BOLD: BEST VALUE IN EACH COLUMN AND CONDITION; UNDERLINED: BEST LEARNED VARIANT WHERE A RULE-BASED SCHEME IS BEST; SHADED: FULL AGENTIC STACK (M5). THRESHOLD-DUP ATTAINS THE LOWEST HANDOVER COUNTS ONLY BECAUSE IT RARELY HANDS OVER, AT THE COST OF THE HIGHEST OUTAGE.

	Method	Outage ↓	Deficit ↓	HO exec./epoch ↓	Ping-pong ↓	Resource ($\times 10^{-3}$) ↓	Common reward ↑
Nominal	M1	$\mathbf{0.131 \pm 0.008}$	$\mathbf{0.065 \pm 0.004}$	0.367 ± 0.082	0.118 ± 0.028	5.48 ± 2.11	$\mathbf{-0.066 \pm 0.005}$
	M2	0.144 ± 0.021	0.072 ± 0.009	0.288 ± 0.067	0.060 ± 0.022	6.65 ± 3.77	-0.073 ± 0.010
	M3	0.149 ± 0.004	0.074 ± 0.002	0.317 ± 0.085	0.061 ± 0.017	6.24 ± 1.36	-0.076 ± 0.002
	M4	0.143 ± 0.013	0.071 ± 0.005	0.345 ± 0.115	0.075 ± 0.039	7.70 ± 4.83	-0.073 ± 0.006
	M5	0.137 ± 0.006	0.069 ± 0.002	$\underline{0.224 \pm 0.046}$	$\underline{0.035 \pm 0.007}$	$\underline{4.35 \pm 1.19}$	-0.070 ± 0.003
	CHO (A3/TTT)	0.345 ± 0.107	0.145 ± 0.042	0.363 ± 0.015	0.172 ± 0.006	$\mathbf{2.64 \pm 0.09}$	-0.146 ± 0.042
	CHO (A3/TTT)+Dup	0.346 ± 0.106	0.146 ± 0.042	0.364 ± 0.016	0.173 ± 0.006	3.13 ± 0.20	-0.147 ± 0.042
	Max-SINR	0.322 ± 0.096	0.137 ± 0.039	0.515 ± 0.013	0.337 ± 0.022	3.30 ± 0.07	-0.138 ± 0.039
	Always-Dup	0.390 ± 0.096	0.168 ± 0.040	0.690 ± 0.007	0.408 ± 0.020	171.89 ± 4.39	-0.211 ± 0.039
	Threshold-Dup	0.429 ± 0.093	0.190 ± 0.042	$\mathbf{0.089 \pm 0.006}$	$\mathbf{0.008 \pm 0.002}$	9.01 ± 1.04	-0.192 ± 0.042
Stress	M1	$\mathbf{0.194 \pm 0.007}$	$\mathbf{0.098 \pm 0.004}$	0.384 ± 0.080	0.117 ± 0.028	5.75 ± 2.28	$\mathbf{-0.100 \pm 0.004}$
	M2	0.204 ± 0.020	0.105 ± 0.009	0.324 ± 0.078	0.066 ± 0.023	6.90 ± 3.52	-0.107 ± 0.010
	M3	0.208 ± 0.007	0.107 ± 0.002	0.352 ± 0.079	0.069 ± 0.012	6.48 ± 0.86	-0.109 ± 0.002
	M4	0.202 ± 0.010	0.104 ± 0.004	0.367 ± 0.116	0.075 ± 0.037	7.79 ± 4.84	-0.106 ± 0.005
	M5	0.199 ± 0.007	0.103 ± 0.003	$\underline{0.255 \pm 0.039}$	$\underline{0.040 \pm 0.009}$	$\underline{4.82 \pm 1.45}$	-0.105 ± 0.004
	CHO (A3/TTT)	0.369 ± 0.077	0.171 ± 0.033	0.370 ± 0.009	0.163 ± 0.011	$\mathbf{2.68 \pm 0.05}$	-0.172 ± 0.033
	CHO (A3/TTT)+Dup	0.373 ± 0.082	0.173 ± 0.035	0.370 ± 0.012	0.163 ± 0.011	3.94 ± 0.24	-0.173 ± 0.035
	Max-SINR	0.352 ± 0.077	0.164 ± 0.033	0.531 ± 0.013	0.328 ± 0.011	3.38 ± 0.07	-0.165 ± 0.033
	Always-Dup	0.414 ± 0.087	0.195 ± 0.037	0.702 ± 0.007	0.395 ± 0.016	168.22 ± 4.26	-0.237 ± 0.036
	Threshold-Dup	0.457 ± 0.075	0.220 ± 0.034	$\mathbf{0.101 \pm 0.004}$	$\mathbf{0.007 \pm 0.002}$	17.64 ± 1.30	-0.224 ± 0.034

and yields the highest outage ratio of all methods. Compared with MAPPO (M1), M5 attains a statistically indistinguishable reliability with a 70% lower ping-pong ratio (Section VII-C), so its reliability gain over the rules is not obtained at the cost of mobility stability. The advantage is not an artefact of averaging: all 125 policy–scenario pairs (25 trained policies $\times$ 5 evaluation seeds) have a lower outage ratio than CHO in the same scenario, in both conditions. The wide CIs of the baselines stem from differences in scenario difficulty; in paired terms, the ratio between the outage of M5 and that of CHO stays within 0.38–0.42 (nominal) and 0.52–0.56 (stress) across the evaluation seeds. The price of this reliability is a moderately higher resource cost: CHO almost never duplicates and thus spends 2.6×10^{-3} versus 4.3×10^{-3} for M5. Adding a duplication trigger to CHO barely changes its behavior (duplication is active in 0.06% of the time), which indicates that reliable use of duplication requires anticipating when a second leg is worth its cost, rather than a fixed SINR threshold.

C. Capability-Structured Comparison of M1–M5

Fig. 6 summarizes the capability ablation. The capability ladder is not intended to produce a monotonic ranking. Among the learned policies, M1 and M5 form two distinct operating points (Fig. 5): their outage ratios are statistically indistinguishable, whereas M5 executes significantly fewer handovers and has a significantly lower ping-pong ratio. On average, M5 also improves every metric in Table IV relative to M2–M4, although only part of these differences are statistically significant with five policies per variant.

Among the learned variants, reliability is similar, and the capabilities mainly change *how* the agents achieve it (Fig. 5 and Table IV). The memoryless M1 attains the lowest outage ratio (0.131 ± 0.008) and the best common reward, but it does so with the most mobility actions. *Memory* (M2) halves the ping-pong ratio (-49%) and reduces handover executions by 21% relative to M1, at the price of a 1.3 percentage-point (pp) higher outage ratio, which is consistent with the agent learning to ride out transient degradations instead of reacting to every telemetry fluctuation. *Reasoning* (M3) yields no measurable gain over M2 (outage $+0.5$ pp, handover executions $+10\%$), although it has the smallest variance of the outage ratio across training seeds. Adding *coordination* alone (M3→M4) gives mixed results: the outage ratio decreases slightly, but handover executions and resource cost increase, and the variance across training seeds grows.

The largest change occurs when *goal regulation* is added (M4→M5): handover executions decrease by 35% and the ping-pong ratio by 53% (both significant), while the resource cost (-44%), the number of failed handovers (-32%), and the outage ratio also decrease on average but not significantly with five policies. At the same time, the variability across independently trained policies shrinks markedly: the CI half-width of the handover executions drops from 0.115 to 0.046 and that of the ping-pong ratio from 0.039 to 0.007. As a result, the full stack operates at a distinct point of the reliability–mobility trade-off (Fig. 5): compared with M1, M5 performs 39% fewer handover executions and has a 70% lower ping-pong ratio (both significant), whereas its outage ratio is 0.6 pp higher, a difference that is not significant. The resource cost (-21%) and the number of failed handovers (-19%) are also lower on average, but these differences are not significant. The same pattern holds under compound stress: handover executions and the ping-pong ratio decrease

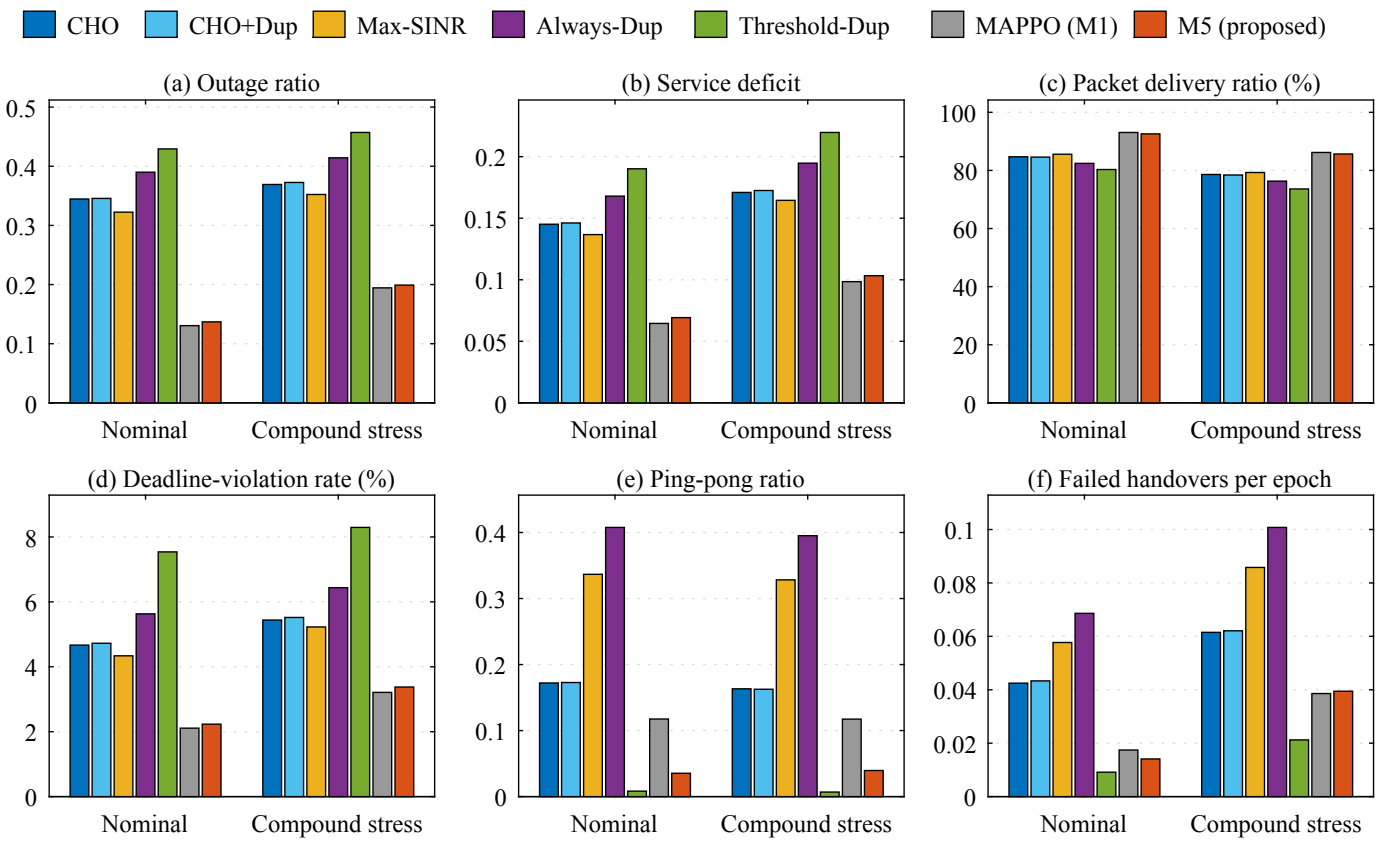


Fig. 4. Comparison of M5 with the five rule-based schemes and with the MAPPO policy M1 under nominal and compound-stress conditions (means over five trained policies for learned methods and over five evaluation seeds for rule-based schemes; 95% CIs are given in Tables IV and VIII). Threshold-Dup attains low ping-pong and handover-failure counts only because it rarely hands over.

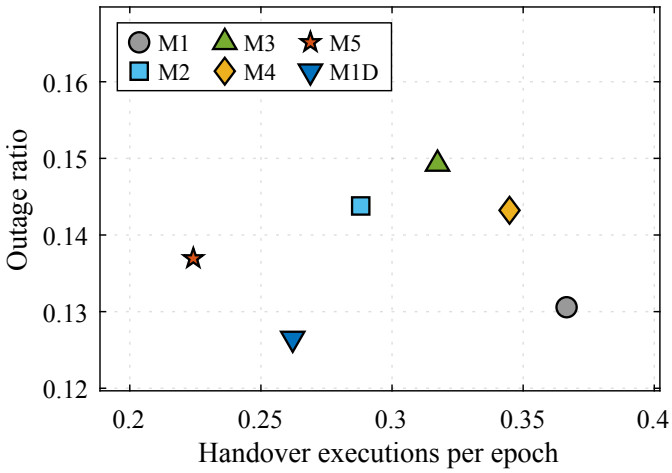


Fig. 5. Reliability–mobility trade-off of the learned variants in the nominal condition: handover executions per epoch versus outage ratio (means over five trained policies). 95% CIs and compound-stress results are given in Tables IV and V.

by 34% and 66% (both significant) at a non-significant +0.5 pp outage difference. We emphasize that the M1–M5 ladder is not monotone: the capabilities interact. Which part of the M4→M5 change stems from goal regulation itself is examined next.

D. Is Goal Regulation Alone Sufficient?

Table V compares M1 and M5 with M1D, which adds only the reliability Lagrangian to the memoryless policy. Goal regulation alone already changes the operating point substantially: compared with M1, M1D reduces the ping-pong ratio by 38% (significant) and the handover executions by 28% (not significant with five policies), while its outage ratio (0.126) and common reward remain statistically indistinguishable from those of M1, and all 50 M1D policy–scenario pairs again have a lower outage ratio than CHO. Goal regulation is therefore the most influential single capability, and much of the efficiency gain of M5 over M1 can be attributed to the Lagrangian reward itself. The remaining capabilities of the full stack nevertheless change the operating point further. Relative to M1D, M5 halves the ping-pong ratio again (−51%, significant) and yields markedly narrower confidence intervals for the handover executions and the ping-pong ratio, i.e., more consistent policies across training seeds; at the same time, M5 has a significantly higher outage ratio (+1.0 pp), a lower common reward, and 24% more preparation requests than M1D. The same ranking holds under compound stress, although the outage difference between M5 and M1D is not significant there. Re-weighting the handover cost post hoc confirms this picture: M5 overtakes M1 in common reward only when each handover execution is weighted about 40 times more strongly than in the training reward, and M1D

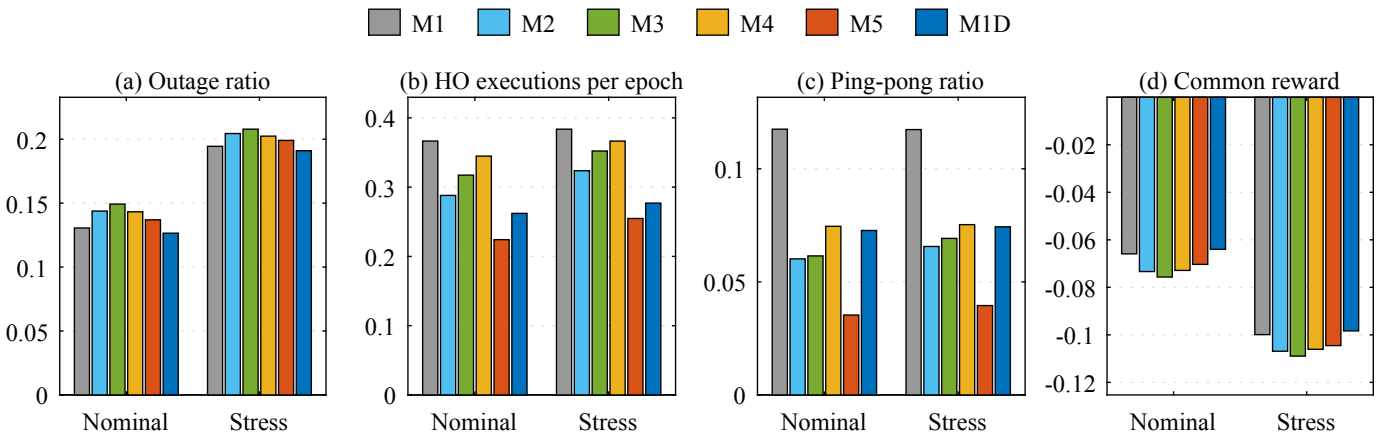


Fig. 6. Capability ablation: the model family M1–M5 and the goal-regulation-only variant M1D under nominal and compound-stress conditions (means over five trained policies; 95% CIs are given in Tables IV and V).

TABLE V
GOAL-REGULATION ABLATION: THE MEMORYLESS POLICY (M1), THE MEMORYLESS POLICY TRAINED WITH THE RELIABILITY LAGRANGIAN (M1D), AND THE FULL AGENTIC STACK (M5) (MEAN ± 95% CI ACROSS 5 TRAINED POLICIES).

	Nominal			Compound stress		
Metric	M1	M1D	**M5**	M1	M1D	**M5**
Outage ratio	0.131 ± 0.008	0.126 ± 0.009	0.137 ± 0.006	0.194 ± 0.007	0.191 ± 0.008	0.199 ± 0.007
Service deficit	0.065 ± 0.004	0.063 ± 0.004	0.069 ± 0.002	0.098 ± 0.004	0.097 ± 0.003	0.103 ± 0.003
HO executions per epoch	0.367 ± 0.082	0.262 ± 0.106	0.224 ± 0.046	0.384 ± 0.080	0.277 ± 0.113	0.255 ± 0.039
Ping-pong ratio	0.118 ± 0.028	0.073 ± 0.037	0.035 ± 0.007	0.117 ± 0.028	0.074 ± 0.034	0.040 ± 0.009
Resource cost ($\times 10^{-3}$)	5.48 ± 2.11	4.95 ± 3.25	4.35 ± 1.19	5.75 ± 2.28	5.35 ± 3.81	4.82 ± 1.45
Common reward	-0.066 ± 0.005	-0.064 ± 0.004	-0.070 ± 0.003	-0.100 ± 0.004	-0.098 ± 0.004	-0.105 ± 0.004
Prep. requests per epoch	3.45 ± 0.95	3.67 ± 0.56	4.56 ± 0.30	3.43 ± 1.00	3.71 ± 0.57	4.51 ± 0.29

attains at least the common reward of M5 for all weights tested. Memory, reasoning, and coordination thus trade a small amount of reliability for further ping-pong suppression and training consistency, rather than being necessary for the efficiency gain.

E. Is the Coordination Signal Used?

Because the variants are trained independently, the M3→M4 comparison alone cannot reveal whether the agents rely on the congestion price. We therefore evaluate each trained M4 and M5 policy a second time on the same episodes with the price forced to zero, which isolates the contribution of the signal within a fixed policy (Table VI). Removing the price increases the handover executions by 18–24%, the ping-pong ratio by 22–35%, and the resource cost by 20–26%, in *every one* of the ten policies and in both conditions. The outage ratio increases in all ten policies in the nominal condition (+5–7%) and in eight of ten under stress, where the effect is small and its CI includes zero. The price is thus an operative input that all ten price-enabled M4/M5 policies have learned to exploit, mainly to suppress unnecessary mobility actions and redundant transmissions. A detailed single-policy diagnostic (training seed 7, 5×200 episodes under compound stress) shows the underlying mechanism: with the price, the agent still prepares candidates but commits more selectively, i.e., the fraction of preparations that end in a switch decreases from 12.8% to 9.4% and more preparations are released without committing. Two caveats apply. First, zeroing the price at evaluation is a distribution shift for a policy trained with it, so the effect measures reliance on the signal rather than the value of training with it. Second, as noted above, adding coordination alone does not guarantee a better operating point; its benefit is consolidated when combined with goal regulation in M5.

F. Is the Reasoning Signal Used?

The M2→M3 comparison showed no measurable gain from reasoning. To test whether trained policies nevertheless rely on the planner, we repeat the within-policy approach of the price ablation and replace the reasoning feature by an uninformative one (Table VII). For M3, the changes are small (at most 3%) and show no consistent degradation; the handover executions even decrease slightly. The M3 actor has thus learned to largely ignore a planner that ranks plans by predicted service continuity alone. For M4 and M5, whose planner additionally prices the incremental resource footprint through (25), removing the reasoning signal increases the handover executions by 7–8%, the ping-pong ratio by 8–10%, and the resource cost by 5–6% in every one of the ten policies, and it lowers the common reward in all ten. Under compound stress, the handover and ping-pong increases persist in all ten policies (+6% and +9–10%), whereas the outage effect becomes small (+1–2%). The reasoner therefore becomes an operative input once it is combined with the congestion price, although its effect is

TABLE VI
WITHIN-POLICY COORDINATION ABLATION: CHANGE CAUSED BY FORCING THE CONGESTION PRICE TO ZERO AT EVALUATION (PRICE-ZERO MINUS NORMAL), MEAN ± 95% CI ACROSS 5 TRAINED POLICIES, RELATIVE CHANGE, AND NUMBER OF POLICIES WITH AN INCREASE.

		M4			**M5**		
	Metric	Δ	Rel.	Incr.	Δ	Rel.	Incr.
Nominal	HO executions per epoch	$+0.061 \pm 0.037$	$+18\%$	5/5	$+0.052 \pm 0.018$	$+24\%$	5/5
	Ping-pong ratio	$+0.027 \pm 0.019$	$+35\%$	5/5	$+0.008 \pm 0.005$	$+22\%$	5/5
	Resource cost ($\times 10^{-3}$)	$+2.20 \pm 2.49$	$+24\%$	5/5	$+1.03 \pm 0.83$	$+23\%$	5/5
	Outage ratio	$+0.0098 \pm 0.0065$	$+7\%$	5/5	$+0.0072 \pm 0.0069$	$+5\%$	5/5
Stress	HO executions per epoch	$+0.063 \pm 0.045$	$+18\%$	5/5	$+0.052 \pm 0.015$	$+20\%$	5/5
	Ping-pong ratio	$+0.026 \pm 0.023$	$+34\%$	5/5	$+0.010 \pm 0.006$	$+26\%$	5/5
	Resource cost ($\times 10^{-3}$)	$+2.27 \pm 2.15$	$+26\%$	5/5	$+1.06 \pm 0.97$	$+20\%$	5/5
	Outage ratio	$+0.0050 \pm 0.0052$	$+2\%$	4/5	$+0.0032 \pm 0.0062$	$+2\%$	4/5

TABLE VII
WITHIN-POLICY REASONING ABLATION IN THE NOMINAL CONDITION: CHANGE CAUSED BY REPLACING THE REASONING FEATURE WITH AN UNINFORMATIVE ONE AT EVALUATION (ABLATED MINUS NORMAL), MEAN ± 95% CI ACROSS 5 TRAINED POLICIES, RELATIVE CHANGE, AND NUMBER OF POLICIES THAT DEGRADE.

	M3			M4			**M5**		
Metric	Δ	Rel.	Degr.	Δ	Rel.	Degr.	Δ	Rel.	Degr.
HO executions per epoch	-0.004 ± 0.003	-1%	0/5	$+0.024 \pm 0.022$	$+7\%$	5/5	$+0.018 \pm 0.006$	$+8\%$	5/5
Ping-pong ratio	-0.002 ± 0.002	-3%	1/5	$+0.009 \pm 0.009$	$+10\%$	5/5	$+0.003 \pm 0.002$	$+8\%$	5/5
Resource cost ($\times 10^{-3}$)	-0.2 ± 0.4	-2%	1/5	$+0.5 \pm 0.5$	$+6\%$	5/5	$+0.2 \pm 0.2$	$+5\%$	5/5
Outage ratio	$+0.001 \pm 0.003$	$+1\%$	4/5	$+0.005 \pm 0.004$	$+4\%$	5/5	$+0.004 \pm 0.004$	$+3\%$	4/5
Common reward	-0.001 ± 0.002	-1%	3/5	-0.002 ± 0.001	-3%	5/5	-0.002 ± 0.002	-3%	5/5

smaller than that of the price itself (Table VI). As for the price ablation, this measures reliance on the signal within a fixed policy rather than the value of training with it.

G. How Do the Agents Use Conditional Handover?

Fig. 7 and Table VIII reveal that the learned agents use the CHO procedure in a qualitatively different way from the rule. The tuned CHO rule stays in 87.8% of its decisions and, once it prepares a target, executes the switch in 73% of the cases; it holds a preparation for 5.3 epochs on average, and 49% of its preparation requests are rejected because many UEs request the same satellite at the same time. The learned agents instead treat a preparation as a short-lived *option*. M5 issues a preparation in 38% of its decisions and releases most of them, so that only 6.3% of the preparations end in a switch; a preparation is held for only 1.3 epochs, i.e., little longer than the preparation timer. This prepare-broadly, commit-selectively behavior is progressively reinforced along the capability ladder (prepare-to-commit ratio of 18.9% for M1, 9.7–13.5% for M2–M4, and 6.3% for M5) and is accompanied by fewer admission conflicts: the preparation rejection rate falls from 35% (M1) to 20% (M5), compared with 49% for CHO, which is consistent with the agents learning to avoid contended targets. PDCP duplication, in contrast, is used very sparingly by all learned variants (active 0.2–0.6% of the time), whereas activating it indiscriminately (Always-Dup, 22.8%) degrades every metric; under the present cost weights, the agents learn that the resource cost of a second leg is rarely justified. The prepare-and-release behavior comes at a price in control-plane signalling: M5 issues 4.6 preparation requests per epoch, compared with 1.0 for CHO. Since the reward charges the reserved resources but not the number of signalling messages, this overhead is not penalized during training and is discussed as a limitation in Section VIII.

H. Service, Satellite Load, and Signalling

The reliability advantage translates directly into delivered traffic (Table VIII). The learned variants deliver 92.1–93.1% of the arriving packets in the nominal condition and 85.3–86.2% under compound stress, compared with 84.7% and 78.6% for CHO; deadline violations and queue overflows are roughly halved (2.1–2.4% vs. 4.7% and 4.5–5.2% vs. 10.3%, respectively). A failed switch is also about half as likely per execution (4.7–6.2% vs. 11.6%). The learned agents carry more traffic—the mean satellite utilization is about 50% versus 36% for CHO—and nevertheless lower the mean of the per-epoch peak requested pressure from 1.35 to 1.15–1.20 and reduce the number of overloaded satellites, i.e., they spread the load more evenly across the constellation slice. Under compound stress, the peak-pressure advantage persists (1.23–1.29 vs. 1.43), whereas the number of overloaded satellites becomes similar to CHO because the disturbances force overload regardless of the association policy.

I. Robustness to Compound Stress

Under the compound-stress pattern, whose traffic-burst and blockage severities lie outside the training range, all learned variants remain far more reliable than the rule-based schemes (Table IV), and the efficiency advantages of M5 are preserved. However, the outage ratio of the learned variants increases by 5.9–6.4 pp, compared with 2.5 pp for CHO and 3.0 pp for Max-SINR. The learned policies thus retain a large absolute

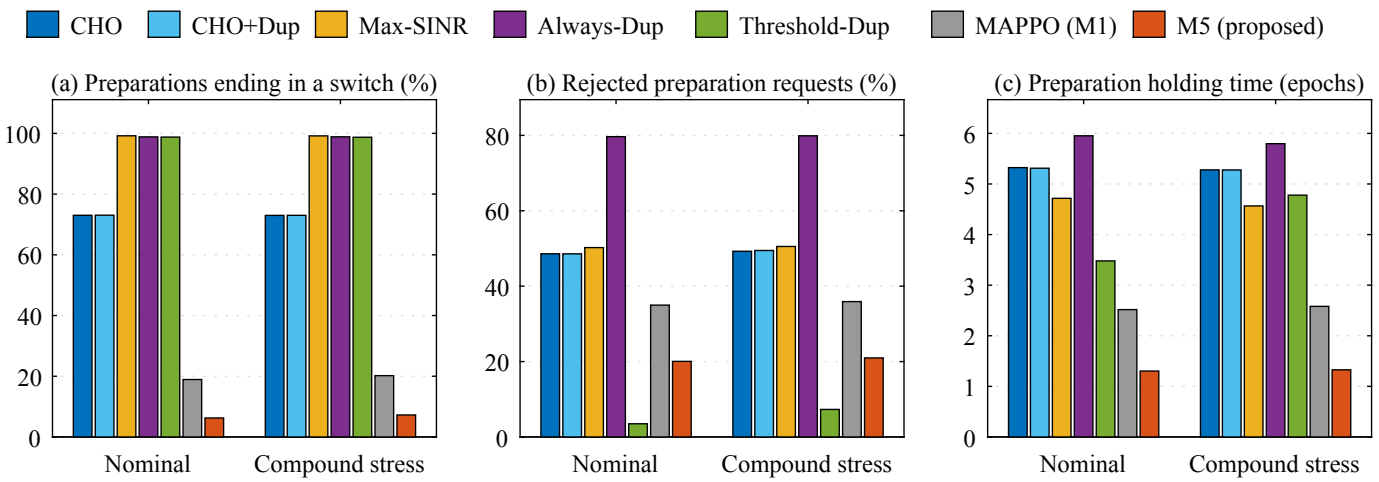


Fig. 7. Use of CHO preparations by the rule-based schemes, MAPPO (M1), and M5 under nominal and compound-stress conditions. Further behavioral metrics are given in Table VIII.

TABLE VIII
EXTENDED METRICS UNDER NOMINAL CONDITIONS (MEANS OVER FIVE TRAINED POLICIES FOR LEARNED VARIANTS AND OVER FIVE EVALUATION SEEDS FOR RULE-BASED SCHEMES; CIS OMITTED FOR READABILITY). COUNTS PER EPOCH ARE SUMMED OVER THE $N = 12$ UES OR $L = 6$ SATELLITES. BOLD: BEST VALUE FOR METRICS WITH A CLEAR PREFERENCE DIRECTION (BEHAVIORAL DESCRIPTORS SUCH AS PREP.→COMMIT, HOLDING TIME, DUPLICATION TIME, AND UTILIZATION ARE NOT RANKED); SHADED: M5.

	Metric	M1	M2	M3	M4	**M5**	CHO	Max-SINR	Al.-Dup	Thr.-Dup
Service	Packet delivery ratio (%)	**93.1**	92.3	92.1	92.4	92.6	84.7	85.6	82.4	80.3
	Deadline-violation rate (%)	**2.11**	2.31	2.38	2.27	2.23	4.67	4.34	5.63	7.54
	Queue-overflow ratio (%)	**4.5**	5.1	5.2	5.0	4.9	10.3	9.8	11.6	11.8
Mobility	HO failures per execution (%)	**4.7**	6.0	6.2	5.9	6.2	11.6	11.2	9.9	10.1
Control plane	Prep. requests per epoch	3.45	4.04	3.68	3.83	4.56	0.97	1.04	3.43	**0.09**
	Prep. rejection (%)	35.0	24.6	25.7	28.1	20.1	48.6	50.2	79.7	**3.5**
	Prep. → commit (%)	18.9	9.7	11.7	13.5	6.3	73.0	99.2	98.9	98.8
	Prep. holding time (epochs)	2.52	1.62	1.84	1.82	1.30	5.32	4.72	5.95	3.48
Duplication	Duplication active time (%)	0.26	0.45	0.39	0.55	0.22	0.00	0.00	22.83	1.12
Satellite load	Mean utilization (%)	49.5	50.1	50.1	49.9	50.6	36.0	38.0	50.9	40.6
	Mean peak requested pressure	**1.15**	1.18	1.20	1.18	1.17	1.35	1.33	1.44	1.56
	Overloaded satellites per epoch	**0.88**	0.98	1.02	0.98	0.98	1.12	1.14	1.58	1.38

TABLE IX
DECISION LATENCY PER CONTROL EPOCH FOR ALL $N = 12$ UES (SINGLE CPU THREAD, UNOPTIMIZED PYTHON/NUMPY; MEAN VALUES).

Model	Actor (ms)	Reasoner (ms)	Total (ms)
M1	0.24	–	0.24
M2	0.41	–	0.41
M3	0.43	19.6	20.1
M4	0.44	21.0	21.5
M5	0.43	21.3	21.7

advantage but are more sensitive to out-of-distribution disturbances than the rules, and none of the capabilities provides a specific robustness gain.

J. Computational Cost

Table IX reports the decision latency. The actor requires less than 0.5 ms per epoch for all UEs, and the analytical reasoner of M3–M5 dominates with about 20 ms, i.e., below 2 ms per UE, which is negligible compared with the 1 s control epoch even for an unoptimized single-threaded implementation.

VIII. DISCUSSION AND LIMITATIONS

Reliability target. Although M5 and M1D regulate the outage pressure during training, their outage ratios on the held-out scenarios (0.137 and 0.126) remain above the threshold $C_{\mathrm{thr}} = 0.10$, as do those of all other variants. In addition, changing the frozen multiplier at evaluation hardly changes the behavior of M5. The dual controller should therefore be understood as a training-time regulator rather than as a goal-conditioned policy that adapts its operating point on demand; enforcing the constraint on unseen scenarios and training with randomized multipliers are open problems.

Reasoning. The ephemeris-based reasoner did not improve performance when it was added alone (M2→M3), and the within-policy ablation shows that M3 ignores it; only the price-aware planner of M4 and M5 is used by the trained policies. Several implementation choices may explain why the purely geometric lookahead is not informative enough. First, the planner extrapolates delayed and noisy radio telemetry. Second, its hypothetical protocol rollout treats a feasible handover completion deterministically, whereas the actual environment

resolves the switch stochastically according to (12). Third, the single-UE hypothetical resource vector accounts for the data-plane preparation reservation but not the explicit control-slot contention term m_ℓ/M; the latter reaches the planner only indirectly through the congestion price. Finally, the recurrent memory may already capture part of the temporal structure that the lookahead attempts to expose. These observations suggest that using the planner as an action prior, incorporating an expected handover-success model, or explicitly representing preparation contention are natural directions for future study.

Mechanism of goal regulation. The M1D ablation localizes much of the efficiency gain in the Lagrangian reward itself, which reduces mobility actions without a loss of reliability even for the memoryless policy. Why this reward suppresses unnecessary handovers, and why combining it with memory, reasoning, and coordination further reduces ping-pong and training variability at a small reliability cost, remain open questions that deserve further study.

Signalling and duplication. The reward charges reserved preparation resources but not the number of preparation messages. The learned prepare-and-release behavior therefore generates 3.5–4.7 times more preparation requests than the CHO rule, and the full stack issues 24% more requests than M1D. A message-level signalling cost should be added to confirm that the observed advantages persist when control signalling is explicitly priced. Moreover, the learned agents activate PDCP duplication only rarely under the present cost weights; the regimes in which duplication becomes worthwhile (e.g., stricter reliability classes or lower duplication cost) remain to be characterized.

Model scope. The study uses a stylized constellation slice with six satellites and twelve UEs, a fixed interference floor, a system-level protocol abstraction, and a fixed scheduler. Scalability to larger constellations and UE populations, explicit interference, and message-level protocol effects remain to be evaluated. Finally, the statistics rely on five independently trained policies per variant.

IX. Conclusion

We formulated the joint control of CHO preparation, PDCP duplication, and primary-satellite switching in a shared-capacity LEO network as a constrained Dec-POMDP and proposed a capability-structured agentic framework that instantiates perception, memory, reasoning, and coordination through lightweight, numerical mechanisms matched to LEO mobility, together with a training-time goal regulator. Across 25 independently trained policies, every learned policy–scenario pair achieved lower outage than the tuned 3GPP-inspired CHO rule. For M5, the mean outage reduction was 60% under nominal conditions and 46% under compound stress. The full capability stack achieved reliability close to that of the simplest learned policy while reducing handover executions by 39% and the ping-pong ratio by 70% under nominal conditions, with markedly smaller variability across trained policies. Targeted ablations showed that goal regulation is the most influential single capability, as it preserves reliability while suppressing mobility actions even for the memoryless policy, and that the trained M4 and M5 policies rely on both the congestion price and the price-aware reasoner. A behavioral analysis showed that the agents use CHO preparations as short-lived options and commit only a small fraction of them, which reduces admission conflicts, raises the packet delivery ratio by about eight percentage points relative to the rule, and lowers the mean per-epoch peak requested pressure. The results also delimit the approach: the geometric reasoner helped only in combination with coordination, the reliability target was not met on unseen scenarios, the learned policies were more sensitive to compound disturbances than the rules, and their prepare-and-release behavior increases control-plane signalling, which the present reward does not price. These findings indicate that goal regulation carries most of the benefit and that the other capabilities are effective mainly in combination, and they point to goal-conditioned regulation and planner-guided action selection as the most promising directions.

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